Department of the Interior
U.S. Geological Survey

# Landsat
# Multi-Satellite Operations Center (LMOC) Requirements Document

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science for a changing world

# Executive Summary

This Landsat Mission Operations (LMO) – Landsat Multi-Satellite Operations Center (LMOC) Requirements Document (LMOCRD) contains the Level 4 requirements to support a Multi-Satellite Operations Center for Landsat 9 (L9) and Landsat 8 (L8).

Landsat represents the world's longest continuously acquired collection of space-based moderate-resolution land remote sensing data. More than four decades of imagery provides a unique resource for those who work in agriculture, geology, forestry, regional planning, education, mapping, and global change research. Landsat images are also invaluable for emergency response and disaster relief.

This document is under Landsat Ground System (GS) Configuration Control Board (CCB) control. Please submit changes to this document, as well as supportive material justifying the proposed changes, via a Change Request (CR) to the Process and Change Management Tool.

# Contents

# Section 1 Introduction

The Landsat mission is a joint mission formulated, implemented, and operated by the National Aeronautics and Space Administration (NASA) and the Department of the Interior (DOI) U.S. Geological Survey (USGS). Landsat is a remote sensing satellite mission providing coverage of the Earth's land surfaces. The Landsat series of satellites continue the 40+ years of global data collection and distribution.

## 1.1 Background

The goal of Landsat is to continue the collection, archival, and distribution of multispectral imagery affording global, synoptic, and repetitive coverage of the Earth's land surfaces at a scale where natural and human-induced changes can be detected, differentiated, characterized, and monitored over time. The Landsat programmatic goals are stated in the United States Code, Title 15 Chapter 82 "Land Remote Sensing Policy" (derived from the Land Remote Sensing Policy Act of 1992). This policy requires that the Landsat Project provide data into the future that are sufficiently consistent with previous Landsat data to allow the detection and quantitative characterization of changes in or on the surface of the Earth. The highly successful Landsat series of missions have provided satellite coverage of the Earth's continental surfaces since 1972. The data from these missions constitute the longest continuous record of Earth's surface as seen from space.

## 1.2 Purpose and Scope

The Landsat Multi-Satellite Operations Center Requirements Document (LMOCRD) establishes the Level 4 requirement set for the Landsat Mission Operations (LMO) – Landsat Multi-Satellite Operation Center (LMOC) supporting both Landsat 8 (L8) and Landsat 9 (L9). This requirement set encompasses the LMOC functionality, performance specifications, and interfaces required to support both missions and serves to verify both Missions Operations functionality in the LMOC architecture.

## 1.3 Document Organization

This document contains the following sections:

- Section 1 introduces the Landsat missions
- Section 2 provides a high-level overview of LMOC
- Section 3 provides the Level 4 LMOC requirements
- Appendix A provides an acronyms list

## 1.4 Terminology

A **requirement** is a specification of a function, capability, interface, or constraint with which the system design must be compliant, verifiable, and have a proven achievement during the mission.

A **shall** statement designates a mandatory requirement the Contractor must comply with. Any deviations from these contractually imposed, mandatory requirements require the approval of the Contracting Officer.

A **mission** attribute for a requirement states the applicability of that requirement toward the particular mission(s).

A **Unique Scene ID** is the term used to scope the L8 Orbit & Path/Row and L9 Scene ID values for image scheduling.

Command and Data Handling **(C&DH) Memory** is the term for the local, non-mass storage space for stored housekeeping telemetry.

**Observatory Mass Storage** is the term used for the mass storage onboard the observatory such as the Solid State Recorder (SSR).

**Observatory On-Board Data Storage** is the term used to cover both the Observatory Mass Storage and C&DH Memory.

A **Collection Request** is the term used to identify either and/or both a Special Collection Request (SCR) and a Calibration Collection Request (CalCR).

An **LTAP Collection Request** is the term used to identify the Landsat 9 Long-Term Acquisition Plan (LTAP) base image acquisition request, not to be confused with a Collection Request.

# Section 2 Landsat Multi-Satellite Operations Center (LMOC) Overview

The LMOC handles management of image and calibration acquisitions, commands the generation for all Observatory activities, and monitors Observatory health, safety, and performance. The LMOC comprises the facility and systems that include the capabilities identified in the following sections.

## 2.1 Telemetry & Command (T&C)

Galaxy is used to provide archive and acquisition of real-time S-Band telemetry from the various antenna complexes, generate and encrypt commands for uplink, manage the Solid State Recorder (SSR), and execute time-based automation tasks.

## 2.2 Data Management (DM)

Mercury manages file distribution and event-based automation within the LMOC. A central agent administers configurations and provides web portal services while various remote agents perform actual automation work.

## 2.3 Trending & Analysis (T&A)

Stars ingests S-Band real-time, S-Band Stored State of Health (SSOH), and X-Band SSOH data; Stars also generates statistical data. It produces both automated and user interface outputs in plot and various machine-readable formats.

## 2.4 Flight Dynamics (FDS)

FreeFlyer, FreeFlyer Meridian, and Multi-Mission Three-Axis Stabilized Spacecraft (MTASS) are used to perform orbit determination and propagation, orbit maintenance, attitude prediction, attitude determination, and Onboard Computer (OBC) sanity checks.

## 2.5 Mission Planning (MPS)

Collection Planning and Analysis Workstation (CPAW) and Systems Toolkit (STK) Scheduler process input from flight dynamics, various collection requests from Data Processing and Archive System (DPAS), and other sources to generate the contact schedules and activity plans for the spacecraft and ground system.

## 2.6 Alerts and Notifications (ANS)

Graylog and What's Up Gold process log messages in order to generate alert notifications to the Flight Operations Team (FOT) Engineers.

## 2.7 Analytical Modeler (AM)

The Joint Activity Simulation for Observatories and Networks (JASON) models mission behavior in a multi-satellite system, including expected spacecraft and ground system activities and realistic failure scenarios.

## 2.8 Infrastructure (Inf)

An amalgamation of Commercial Off-the-Shelf (COTS) and open source software, which provides services related to network, compute, disk, authentication, remote access, security, backups, video distribution, and other tools.

# Section 3 LMOC Level 4 Requirements

## 3.1 General

**LMOC-2424:** The LMOC shall support a mission life of at least 15 years.
**Rationale:** The Project Mission goal expresses the need for a scalable system since the LMOC will need to support observatory ops through decommissioning.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2413:** The LMOC shall have an availability of 99.9% for observatory telemetry and commanding operations.
**Rationale:** To ensure command and telemetry system along with the networking within the LMOC, is available over the period of a month, to maintain the health and safety of the Observatory.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2416:** The LMOC shall ensure protection exists from a single point of failure within the critical telemetry and command functions.
**Rationale:** The T&C system must have no single point of failures that will preclude observatory health and safety.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2415:** The LMOC shall have a mean time to restore telemetry and commanding operations within 1 minute or less.
**Rationale:** This provides for a timely transfer of command authority and telemetry monitoring from one system to another in the event of a failure.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2417:** The LMOC shall have a combined availability of 98.5% for all non-telemetry and commanding subsystems.
**Rationale:** To ensure all other LMOC subsystems are available without the stringent constraints of the T&C system. Availability is to be calculated as total downtime per month on a per-subsystem basis.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2427:** The LMOC shall be able to simultaneously operate the observatory and observatory simulators without impact to mission operations.
**Rationale:** The lMOC must be able to verify and validate activities with observatory simulators (e.g. LSIMSS, Softsim/Softbench, and SOS) while mission operations are on-going.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5189:** The LMOC shall be able to add and remove systems with no loss in data or interruption to operations.
**Rationale:** The real-time system is expected to be fault-tolerant of the loss of operational, backup or off-line systems. This drives the expectation that all subsystems

will be modular and scalable in design, with a warm backup plug and play architecture so that the redundant copy of that subsystem is available for user initiation on demand.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3530:** The LMOC shall make available to users one orbit's worth of stored housekeeping data within 30 minutes of receipt of the data from GNE.
**Rationale:** The full end to end transfer, processing, display of stored telemetry needs to be completed in a timely manner for all operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3442:** The LMOC shall generate a conflict-free stored command load within 2 hours, including user interaction time, starting from the beginning of the scheduling process.
**Rationale:** Supports fast turnaround for operations efficiency, anomaly response, and short-notice emergency requests.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2422:** The LMOC shall support a single 8-hour by 5-day shift (M-F) approach and operate autonomously whenever not staffed.
**Rationale:** LMOC automation required to support a minimal staffing level for nominal operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2423:** The LMOC shall have the ability to perform nominal operations autonomously for at least 72 hours.
**Rationale:** The LMOC needs to support nominal operations autonomously during a 3-day weekend at a minimum.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3448:** The LMOC automation functions shall be capable of being enabled or disabled.
**Rationale:** All automation should be capable of being disabled to support ongoing operations as required. Manual activities are needed during L&EO, commissioning and during anomalous conditions.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2428:** The LMOC shall allow execution of operations both autonomously and by user input through a command line interface.
**Rationale:** Provides capability to execute scripts to perform LMOC activities.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5122:** The LMOC shall have a test environment to allow for verification of changes before they are operationally implemented.
**Rationale:** The Test Environment is needed to test software, hardware, operating system, drivers, application, scripts, procs and operational concept changes for each sub-system so they don' t impact operational systems.

**Mission:** Landsat 8 and Landsat 9

**LMOC-2426:** The LMOC shall support training, testing, and system maintenance while meeting all requirements.
**Rationale:** LMOC training, testing and systems maintenance activities shall not impact on-going operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5142:** The LMOC shall tag all operational products by Spacecraft ID.
**Rationale:** Allows LMOC personnel and subsystems to correlate simulated and spacecraft data by Spacecraft ID to delineate between Missions.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2418:** The LMOC shall time tag all observatory and ground system data.
**Rationale:** Allows operator to correlate ground system and spacecraft data output displays & reports.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2410:** The LMOC shall use Universal Time Coordinated (UTC) for all systems, displays and products.
**Rationale:** Utilize standard time reference across all LMOC systems.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2430:** The LMOC shall utilize common time formats across LMOC systems.
**Rationale:** The LMOC needs to utilize a common time format for all operations products generated in the LMOC. Additionally the LMOC systems and displays also need to maintain a consistent time zone and time format's where possible.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2425:** The LMOC shall employ a time stamp resolution of at least one millisecond for all operational products used and stored within the LMOC.
**Rationale:** Define time stamp precision within the LMOC for all operational products.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2421:** The LMOC shall generate and locally retain logs from all LMOC systems.
**Rationale:** Maintaining the record of all system interactions, events is useful for both troubleshooting and supporting operational activities and for IT Security incidents.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5109:** The LMOC shall distinguish between system and user actions in the system logs.
**Rationale:** Distinguishing between system and operator actions is an important part of tracking and optimizing operational activities and for IT Security incidents.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2431:** The LMOC shall provide users the ability to print documents, displays, logs, schedules and operational products.
**Rationale:** Provides the capability to generate hard copies vs. soft copies of various products.
**Mission:** Landsat 8 and Landsat 9

## 3.2 Security

**LMOC-2433:** The LMOC shall employ security controls of NIST PUB 800-53 moderate level system in accordance with corresponding USGS guidance within the LMOC IT Security Control Matrix.
**Rationale:** The LMOC IT security posture must meet NIST PUB 800-53 for a moderate threat level system. The USGS IT Control Matrix adheres to as part of compliance with NIST PUB 800-53.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5140:** The LMOC shall restrict access to LMOC systems to only authorized personnel.
**Rationale:** The LMOC needs the ability to provide users with individual and group policy based access to LMOC Systems, Websites and Applications by using an authentication mechanism (e.g. Active Directory, OpenLDAP).
**Mission:** Landsat 8 and Landsat 9

**LMOC-5199:** The LMOC shall support multifactor authentication for user authentication to LMOC systems.
**Rationale:** LMOC users need multifactor authentication to access consoles, systems, remote access. (e.g. Smart Card, RSA).
**Mission:** Landsat 8 and Landsat 9

**LMOC-2437:** The LMOC shall support the Government-furnished system configuration monitoring tool without impacting mission operations.
**Rationale:** The Government is providing a software tool to consolidate LMOC system configuration information for security oversight. LMOC systems will need to be configured in a way that allows the tool to receive information (e.g. OS version).
**Mission:** Landsat 8 and Landsat 9

**LMOC-2435:** The LMOC shall support system patches and updates without impacting mission operations.
**Rationale:** Security policies require timely patching and updates of systems. The LMOC must be able to handle IT Security activities and ongoing operations simultaneously.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2434:** The LMOC shall support credentialed scans without impacting mission operations.

**Rationale:** Security policies enforce the need to periodically execute scans on operational systems. The LMOC needs to be able to support these while supporting operations and the availability constraints.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2436:** The LMOC shall forward all system logs to the Government-furnished log monitoring system.
**Rationale:** The Government is providing a centralized system for consolidation of LMOC system logs for security oversight for the life of the mission. Therefore LMOC systems will need to interface to it by sending logs.
**Mission:** Landsat 8 and Landsat 9

**LMOC-38943:** The LMOC shall employ security controls of the Department of Homeland Security (DHS) High Value Asset (HVA) Control Overlay in accordance with corresponding USGS guidance within the LMOC IT Security Control Matrix.
**Rationale:** The LMOC IT Security Control Matrix also meet the control enhancements listed in the DHS HVA Control Overlay.
**Mission:** Landsat 8 and Landsat 9

## 3.3 Infrastructure

**LMOC-2442:** The LMOC shall receive a master time reference from the Government provided source.
**Rationale:** Synchronize LMOC system time to common source.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2443:** The LMOC shall synchronize to the Government provided master time reference.
**Rationale:** Maintain time consistency across all LMOC workstations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5188:** The LMOC shall provide a centralized Domain Name System (DNS) to resolve system host names and IP addresses for all systems and devices connected to the LMOC network.
**Rationale:** Use of local, centralized DNS is intended to eliminate dependency on external DNS services, allow simple name resolution and access of LMOC systems and relevant external hosts, and provide a single point of update for operational changes (i.e. allow updates to be resolved quickly in the event of a host failure or other local or remote system name/IP address change).
**Mission:** Landsat 8 and Landsat 9

**LMOC-5206:** The LMOC shall provide a centralized location for the management of LMOC documentation.
**Rationale:** The LMOC needs a central location that can be utilized for archiving and retrieval of approved SOP's, COP's and any additional LMOC and bLMOC specific documentation.

**Mission:** Landsat 8 and Landsat 9

**LMOC-5111:** The LMOC shall provide web based access to the centralized location for LMOC documentation.
**Rationale:** The LMOC needs the ability to access the documentation system remotely via a web based interface, allow the creation of LOP's, COP's (ie. Wiki) and to access any other additional LMOC and bLMOC specific documentation.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5139:** The LMOC shall provide authorized users with policy based access to all LMOC documentation.
**Rationale:** The LMOC needs the ability to provide users with individual or group policy based access to read and write LMOC specific documentation that is stored at the centralized location.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3647:** The LMOC shall provide configuration management of LMOC systems and software.
**Rationale:** Configuration management is needed to prevent unauthorized changes from being applied and affecting the LMOC systems and software. This includes the management of software, documentation and scripts used by operations that handled on a system level which require CM control.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5110:** The LMOC shall provide the ability to rapidly deploy new systems using a pre-existing base image within 30 minutes.
**Rationale:** Operations needs the ability to have pre-configured operating system images with any applicable FOT or Infrastructure subsystem software deployed in a timely fashion, this excludes any rapid deployment of large archives or other hardware specific items.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2444:** The LMOC shall provide an web based event countdown clock that displays upcoming mission events.
**Rationale:** All spacecraft and ground activities need to be displayed, in reference to UTC, in a highly accessible manner for use by operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3627:** The LMOC shall provide authorized users with policy based remote access to the LMOC systems, websites and applications.
**Rationale:** Remote access for authorized users is needed for operations and development. (ex. Engineers and other authorized users (e.g. FOT, CVT, Spacecraft Vendors) need access to real-time and stored observatory telemetry, trending and analysis, engineering data and operational products outside of the LMOC.)
**Mission:** Landsat 8 and Landsat 9

**LMOC-2420:** The LMOC shall support up to 30 concurrent remote users.
**Rationale:** Remote access capabilities for real-time and stored observatory telemetry, trending and analysis, engineering data and operational products access by FOT, spacecraft, instrument, government and support personnel must be robust enough to handle multiple users simultaneously.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5186:** The LMOC shall perform automated backups of all critical systems.
**Rationale:** Critical LMOC systems need to be backed up frequently to ensure no impact to operations occurs if any fault happens to those critical systems.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5187:** The LMOC shall retain all system backups for at least 30 days.
**Rationale:** LMOC backups only need to be kept 30 days as system images older than 30 days may no longer be relevant to operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-6107:** The LMOC shall be able to restore data from all critical system backups.
**Rationale:** LMOC backups need to be utilized during operations to recover whole systems along with individual files within a backup to the imaged system.
**Mission:** Landsat 8 and Landsat 9

## 3.4 Facility

**LMOC-5200:** The LMOC shall be able to accommodate simultaneous activities for both Landsat 8 and Landsat 9 observatories.
**Rationale:** The LMOC facility needs to allow for multiple spacecraft operations to occur at the same time without negatively impacting either mission.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2449:** The LMOC shall support combined Landsat 8 and Landsat 9 operations within an area not to exceed 1200 sq. ft.
**Rationale:** Define LMOC operations room space allocations; this does not include space for production systems. This square footage includes the space lost due to furniture and facilities equipment (ex. 800 sq. ft. usable space).
**Mission:** Landsat 8 and Landsat 9

**LMOC-2450:** The LMOC servers and network equipment for both Landsat 8 and Landsat 9 shall reside within 6, 48-U racks.
**Rationale:** Constrain space needs within the LMOC Equipment Room.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2448:** The LMOC shall utilize redundant uninterruptible power for all LMOC systems.
**Rationale:** Provide power redundancy to prevent localized power interruptions from affecting operations by using different facility feeds and facility Uninterruptable Power

Supplies (UPS). This includes the use of Automatic Transfer Switch (ATS) for suppling power to single Power Supply Units (PSU) devices like workstations and direct redundant Power Distribution Units (PDU) feeds to dual PSU devices like servers. Facility power outages and maintenance occur infrequently which can cause interruptions to LMOC Operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2439:** The LMOC shall operate in a temperature range from 10 degrees C to 29 degrees C.
**Rationale:** Equipment within LMOC to operate within temperature range for LMOC facility
**Mission:** Landsat 8 and Landsat 9

**LMOC-2440:** The LMOC shall operate in a relative humidity range from 40% to 60%.
**Rationale:** Equipment within LMOC to operate within humidity range for LMOC facility.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2441:** The LMOC shall monitor the environmental conditions of the Spacecraft Observatory Simulator's (S/OS) and notify LMOC users if any conditions exceed a user-defined threshold.
**Rationale:** Notify personnel of potential environmental issues such as when temperature, humidity or water within the facility that houses the Landsat 9 and Landsat 8 S/OS's exceed an acceptable threshold. Equipment could be at risk if conditions rise to dangerous levels.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2451:** The LMOC shall provide dual displays for all user accessible workstations.
**Rationale:** Provide display space for LMOC operations
**Mission:** Landsat 8 and Landsat 9

**LMOC-2445:** The LMOC shall provide wall mounted screen displays viewable from any workstation within the LMOC.
**Rationale:** The mounted displays will show the heads-up status such as upcoming passes, trending data, telemetry displays which may be best displayed to the entire LMOC.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2446:** The LMOC shall provide users the ability to configure the information routed to wall mounted displays.
**Rationale:** This includes the use of a multi-in/out matrix switch, wireless keyboards & mice or thin clients to facilitate the mounted displays. Personnel within the LMOC should have the ability to control any computer attached to a wall-mounted display.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5146:** The LMOC shall be able to output a single display of each LMOC subsystem on the wall mounted displays.

**Rationale:** The FOT uses this functionality to share system displays with a large group during key activities, such as during post-launch deployments.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2447:** The LMOC shall provide a black/white and color printing capability within the LMOC.
**Rationale:** Printers are required since hard copy of some items will be necessary.
**Mission:** Landsat 8 and Landsat 9

## 3.5 Mission Planning

### 3.5.1 General Image Planning and Scheduling Requirements

**LMOC-5124:** The LMOC shall support a planning period of 1 to 72 hours.
**Rationale:** The planning period covers all planned activities for all operational concepts (e.g. planning, scheduling and command load generation). By default Landsat 8 uses a 72 hour load, but certain operational constraints such as early testing could result in single-orbit loads
**Mission:** Landsat 8

**LMOC-5232:** The LMOC shall support a planning period of 1 to 96 hours.
**Rationale:** The planning period covers all planned activities for all operational concepts (e.g. planning, scheduling and command load generation). By default Landsat 9 uses a 96 hour load, but certain operational constraints such as early testing could result in single-orbit loads
**Mission:** Landsat 9

**LMOC-6110:** The LMOC shall have the ability to schedule image data collections consisting of any instrument combination.
**Rationale:** Nominally, image data collections include both instruments, but single-instrument collection must not be precluded. For Landsat 9 this means the ability to schedule OLI-2 only, TIRS-2 only, and OLI-2 + TIRS-2 collections; for Landsat 8 it encompasses OLI only, TIRS only, and OLI + TIRS collections.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2457:** The LMOC shall incorporate the WRS-2 model as defined in GSFC 429-02-07 LDCM Worldwide Reference System-2 Memorandum (WRS-2) for image collection planning.
**Rationale:** The LMOC uses the WRS-2 memorandum as the definition and model of the WRS-2 grid, and must build the WRS-2 grid specification into LMOC systems as appropriate in order to maintain orbit and precisely execute imaging acquisition for processing requests and generating operational products (e.g. WRS2TTT).
**Mission:** Landsat 8 and Landsat 9

**LMOC-2458:** The LMOC shall generate and maintain a WRS-2 path/row to time translation table (WRS2TTT) per the DPAS-MOC ICD.

**Rationale:** The WRS2TTT product associates WRS-2 grid traversal with predicted collection time, and as such is pivotal for image collection planning and scheduling activities, and also for later use in reporting across the ground system. The WRS2TTT contains the time at which each path/row in the WRS-2 grid traversed over the reported planning cycle will be at the observatory nadir position, the solar angle for each path/row, and the cross-track error for each path/row (quantifies the deviation, perpendicular to nadir point flight direction, from the WRS-2 compliant position).
**Mission:** Landsat 8 and Landsat 9

**LMOC-2459:** The LMOC shall provide the WRS2TTT to DPAS per the DPAS-MOC ICD.
**Rationale:** The LMOC generates and sends the WRS2TT to the DPAS for input to ground system reporting functions. As an example this can be used to generate maps of capture success rate vs position.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2455:** The LMOC shall generate and display a graphical report of all scheduled collection activities occurring within a user-selectable time period.
**Rationale:** A graphical report enables a fast and intuitive check for certain activities or activity constraints, which are often related to positions within an orbit or an area such as the SAA.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2456:** The LMOC shall provide users the ability to display scheduled collection activities in tabular formats.
**Rationale:** Due to the volume of individual scenes, these reports can get quite large. Tabular formats like CSV or XML make it easier for staff to ingest the data into other tools for assessment, producing maps, matching to database records, etc.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5204:** The LMOC shall provide users the ability to generate human readable reports on all mission planning products.
**Rationale:** A human readable format enables quality assurance, quality control, and troubleshooting.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2466:** The LMOC shall provide mission planning products to the controlled repository.
**Rationale:** Needed for a controlled source where mission planning products are available and may be retrieved.
**Mission:** Landsat 8 and Landsat 9

### 3.5.2 Collection Requests

**LMOC-2469:** The LMOC shall receive Special Collection Requests (SCR) from DPAS per the DPAS-LMOC ICD.

**Rationale:** The LMOC planning & scheduling function receives special collection requests for imaging acquisitions other than LTAP Collection Requests (e.g. coastal scene requests, non-standing night scene requests, and ascending daylight scenes) from DPAS.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2468:** The LMOC shall provide users the ability to generate collection requests, on demand, per the DPAS-LMOC ICD.
**Rationale:** Though not expected to be a nominal use case, this capability is needed to generate local, ad-hoc image data collections (e.g. nominal earth imagery in the absence of LTAP Collection Request input from DPAS, off-nadir imagery, night imaging, ascending daylight scenes, calibration intervals) used primarily for testing, pre-launch, and LEO operations when the DPAS-LMOC interface for collection requests and LTAP Collection Request transfer is either not used or not available.
**Mission:** Landsat 8 and Landsat 9

**LMOC-35452:** The LMOC shall be capable of scheduling selective scene collection and down-link for IC stations.
**Rationale:** Some IC stations do not request all scenes which could potentially be downlinked during a Landsat 8/9 overfly. Therefore, the LMOC needs to be able to specify which scene can be scheduled for real-time downlink and reported on a per-IC station basis. These scenes may not be a contiguous interval.
**Mission:** Landsat 8 and Landsat 9

**LMOC-33874:** The LMOC shall track and report images not specified by a collection request with unique identifier.
**Rationale:** In order to avoid data loss, the LMOC sometimes commands the spacecraft to continue recording when there is a short gap between intervals. DPAS would like to receive status of these images in the database as "flywheel" collects, as well as make them available to the science community. As such, the LMOC must create unique IDs and include them in the IDSR.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5246:** The LMOC shall reference the WRS-2 coordinate system for all scene collection requests and reporting when the observatory is not flying in a WRS-2-compliant orbit.
**Rationale:** All calibration sites, field work, underflights, and other planning activities during Landsat 9 commissioning will reference the long-established WRS-2 grid. The LMOC needs to be capable of translating between WRS-2 grid references and the current projected observatory orbit.
**Mission:** Landsat 9

**LMOC-2470:** The LMOC shall process collection requests into the image collection schedule.
**Rationale:** The collection requests need to be scheduled for execution by the LMOC.
**Mission:** Landsat 8 and Landsat 9

**LMOC-32446:** The LMOC shall receive collection request modifications from DPAS per the DPAS-LMOC ICD.
**Rationale:** SCRs may need to be updated or canceled for any image acquisitions that fall outside of the planning period.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2472:** The LMOC shall process collection requests that modify or cancel existing collection requests.
**Rationale:** Collection requests submitted to the LMOC may need to be changed or deleted.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5325:** The LMOC shall include collection requests received at least 24 hours prior to observatory over-flight of the requested scene(s) for consideration within the image collection schedule during LEO and commissioning.
**Rationale:** LMOC will plan to acquire special requests and non-routine calibration activities within a reasonable time frame that don't violate constraints outside of emergency requests.
**Mission:** Landsat 9

**LMOC-5137:** The LMOC shall include collection requests received at least 72 hours prior to observatory over-flight of the requested scene(s) for consideration within the image collection schedule.
**Rationale:** LMOC will plan to acquire special requests and non-routine calibration activities within a reasonable time frame that don't violate constraints outside of emergency requests.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2471:** The LMOC shall manage up to one year of active collection requests.
**Rationale:** Landsat data coverage is seasonal, so requests may span up to a year into the future. This requirement is intended to help scope the size of relevant data storage (table sizes, etc.)
**Mission:** Landsat 8 and Landsat 9

**LMOC-2473:** The LMOC shall maintain past collection requests in the system for DPAS-LMOC interface use for no less than 3 weeks.
**Rationale:** The DPAS-LMOC interface includes the ability to status past collection requests. Because of the large potential number of collection requests and to prevent this information from becoming an operational burden on the LMOC, the long term archival of collection request status information is allocated to the DPAS. This requirement represents the constraints on related LMOC information storage so that the system can be properly sized.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2475:** The LMOC shall generate and maintain collection request receipt confirmation per the DPAS-LMOC ICD.
**Rationale:** When activities such as generating new stored command loads change the status of requests, that status change needs to be communicated to DPAS per the DPAS-LMOC ICD.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2474:** The LMOC shall provide collection request receipt confirmation to DPAS per the DPAS-LMOC ICD.
**Rationale:** The LMOC provides persistent collection request status to the DPAS for user confirmation (status may include Success or Failure) to indicate successful ingest of the file or failure that need further action such as resend of the file.
**Mission:** Landsat 8 and Landsat 9

### 3.5.3 Image Data Collection Schedules

**LMOC-3354:** The LMOC shall receive the LTAP Collection Request from DPAS per the DPAS-LMOC ICD.
**Rationale:** The DPAS sends the LTAP Collection Request to the LMOC for use in image collection planning activities.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3355:** The LMOC shall verify the contents and format of the LTAP Collection Request per the DPAS-LMOC ICD.
**Rationale:** The LMOC needs to ensure the delivered content is valid prior to ingesting the data into the mission planning and scheduling function.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3356:** The LMOC shall plan the acquisition of all scenes within the LTAP Collection Request and all collection requests that occur within the planning period.
**Rationale:** The LMOC will plan to acquire all land (Descending Daylight Land), collection requests (Ascending Daylight, Night, Ocean scenes, requested calibrations), and nominal calibration activities for each planning period. For reference, utilizing the DRC-16 analysis, the average number of imagery scenes per day over the 16 day cycle is 738. During constraint checking some scenes may be removed to accommodate higher priority observatory activities.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5115:** The LMOC shall plan routine instrument calibrations within each planning period.
**Rationale:** The LMOC generates plans and schedules for nominal instrument calibrations internally; for calibrations managed by the Cal/Val Team and Mission operations management personnel, the collection request function is utilized.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3357:** The LMOC shall provide the ability to ignore the LTAP Collection Request during image planning.
**Rationale:** The LMOC needs to allow the FOT to enable/disable the LTAP Collection Request within the planning function (i.e. use collection requests only) during L&EO, commissioning, and during anomalous conditions.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3358:** The LMOC shall schedule all planned collection activities that do not violate constraints.
**Rationale:** The LMOC will only eliminate scenes or intervals that violate constraints. Planned collection activities include LTAP Collection Request, collection requests (WRS-2 scene requests and calibration requests), and routine calibration activities.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3361:** The LMOC shall provide a graphical display of activities, both scheduled and not scheduled, within the planning period.
**Rationale:** Provide a quick reference display for activities being performed, rejected or special events for projection onto a LMOC display.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3362:** The LMOC shall generate the Image Data Status Report (IDSR) per the DPAS-LMOC ICD.
**Rationale:** The IDSR contains the consolidated list of all scheduled scene UIDs, their collection request identifiers, interval UID, and related metadata (includes all LTAP Collection Request earth imaging scenes, scene collection requests, requested and routine calibrations, and any other imaging activities with concrete start/stop times and resources (ex. OLI-2 only, TIRS-2 only, OLI-2 + TIRS-2 for Landsat 9 or OLI only, TIRS only, OLI + TIRS for Landsat 8). In addition, it includes the status of the collection requests that are new or updated or cancelled within the past 24 hours. This information is needed to provide status of the requests to the user community.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3363:** The LMOC shall generate a unique scene ID (scene UID) for each collection activity scheduled per the DPAS-LMOC ICD.
**Rationale:** These scene ID values apply to individual scenes within the LTAP Collection Request and collection requests, but also individual calibration activities (i.e. activities in the IDSR); these IDs will be used to associate records across the ground system, so they must be unique for the life of the mission.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3364:** The LMOC shall provide users the ability to request and display any Image Data Status Report (IDSR).
**Rationale:** All schedules and reports should be viewable for reviews, comparisons, or analyses.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3366:** The LMOC shall provide users the ability to generate a Image Data Status Report (IDSR) for a user-specified time interval of up to 16 days.
**Rationale:** Provide the capability to create IDSR for an entire WRS-2 16 day cycle.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3367:** The LMOC shall provide Image Data Status Report (IDSR) to DPAS per the DPAS-LMOC ICD.
**Rationale:** The LMOC provides the IDSR to DPAS for image collection accounting and reporting. These metrics include the complete list of scenes which were scheduled to be collected and a distinct notification of failure to schedule scenes along with a reason / justification. In addition, it includes the status of the collection requests that are new or updated or cancelled within the past 24 hours. This information is needed to provide status of the requests to the user community.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3371:** The LMOC shall generate the IC Scene Transmit Schedule per the DPAS-LMOC ICD.
**Rationale:** The IC Scene Transmit Schedule (STS) identifies the planned acquisitions and downlinks of science data. The LMOC passes the IC Scene Transmit Schedule to the DPAS for distribution to the ICs.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3370:** The LMOC shall provide the IC Scene Transmit Schedule to DPAS per the DPAS-LMOC ICD.
**Rationale:** The IC Scene Transmit Schedule (STS) identifies the planned acquisitions and downlinks of science data. The LMOC passes the station specific IC Scene Transmit Schedule to the DPAS for distribution to the ICs.
**Mission:** Landsat 8 and Landsat 9

### 3.5.4 Interval Planning

**LMOC-3374:** The LMOC shall plan and schedule the observatory imaging intervals.
**Rationale:** LMOC to perform mission image collection activities by combining individual scene collections into continuous imaging intervals.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3376:** The LMOC shall generate and assign a unique identifier for each imaging interval (interval UID) based on an identifier scheme defined in conjunction with the Government.
**Rationale:** Unique identifier used to track imaging intervals throughout the ground system; the interval UIDs are unique for the life of the mission.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3375:** The LMOC shall map image data collection activities to image data collections requests, scene UIDs and interval UIDs.

**Rationale:** This entire mapping will be resolved in the IDSR; the SIFMT will also reference relevant identifiers. This mapping includes all image acquisition activities (including LTAP Collection Request scenes, collection request activities, routine calibrations), regardless if an individual collection request identifier exists for the activity.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5121:** The LMOC shall define and maintain direct mapping between interval UIDs and Root File Names generated for use on the spacecraft.
**Rationale:** This is key mission data management functionality; since root file names generated and uploaded to the spacecraft won't support mission-long uniqueness, the LMOC needs to maintain a tight coupling between interval UIDs and root file names; the SIFMT report will contain the interval UID field along with the root file name field in order to make this important relationship clear to the rest of the Ground System.
**Mission:** Landsat 9

**LMOC-5250:** The LMOC shall define and maintain direct mapping between interval UIDs and Root File IDs generated for use on the spacecraft.
**Rationale:** This is key mission data management functionality; since Root File IDs generated and uploaded to the spacecraft won't support mission-long uniqueness, the LMOC needs to maintain a tight coupling between interval UIDs and Root File IDs; the SIFMT report will contain the interval UID field along with the Root File ID field in order to make this important relationship clear to the rest of the Ground System.
**Mission:** Landsat 8

**LMOC-3377:** The LMOC shall generate and maintain the Scene to Interval File identifier Mapping Table (SIFMT) per the GNE-LMOC ICD.
**Rationale:** The LMOC needs to create the SIFMT to track each scene to the interval and file it is being captured within. This file is used as the basis for tracking image processing through the rest of the Ground System.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3378:** The LMOC shall provide the SIFMT to the GNE per the GNE-LMOC ICD.
**Rationale:** The SIFMT is needed by the GNE for interval assembly. It matches downlinked data with the appropriate sensors and path/row combination. The LMOC passes the SIFMT to the GNE.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3379:** The LMOC shall maintain the collection and downlink status of all scene UIDs and Interval UIDs.
**Rationale:** Need to track status of each scene planned, scheduled, executed, and downlinked.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3380:** The LMOC shall predict instrument calibration opportunities for four weeks in the future.

**Rationale:** This capability is needed to support sensor calibration plans for Lunar and Solar calibration events.
**Mission:** Landsat 8 and Landsat 9

### 3.5.5 Constraint Checking

**LMOC-5114:** The LMOC shall apply solar angle constraints embedded in the LTAP Collection Request and collection requests per the DPAS-LMOC ICD.
**Rationale:** The DAM provides the solar angle at which WRS-2 scenes are collected; the LMOC receives this information from DPAS and applies it as part of the schedule generation process. The generic LTAP Collection Request will have a solar angle constraint which applies to all scenes within the LTAP Collection Request; additional collection requests will be accompanied by a solar angle constraint which is specific to each request.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3398:** The LMOC shall detect when the spacecraft star trackers are occulted by the Earth, Moon, or Sun over the duration of the attitude predict file.
**Rationale:** Report any interference events with the star tracker that may cause potential problems.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3393:** The LMOC shall generate image data collection schedules based on configurable engineering constraints.
**Rationale:** Image collections need to be planned within configurable constraints like boresight to sun angles, duty-cycle of components.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3394:** The LMOC shall provide users the ability to manage resource definitions.
**Rationale:** Resources are used to ensure that activities are not scheduled in conflict with the availability of resources.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3395:** The LMOC shall provide users the ability to manage resource availability data.
**Rationale:** The capability to manage the information regarding the availability of each resource.  The instruments (OLI-2 or TIRS-2) and individual Ground Stations, are examples of a resource.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3396:** The LMOC shall provide users the ability to manage mission resource constraint data.
**Rationale:** Have the ability to add/update/modify various resource constraints in the system.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3397:** The LMOC shall provide users the ability to modify resource constraint thresholds.
**Rationale:** In cases where a resource constraint is specified with maximum and/or minimum values, these values must be changeable using The LMOC software.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3390:** The LMOC shall implement a user-configurable constraint against the maximum data volume allowed in observatory mass storage.
**Rationale:** Observatory mass storage volume constraints may vary depending on phase of mission and other operations-determined factors. The LMOC must always plan image acquisitions and housekeeping data storage within this constraint.
**Mission:** Landsat 8 and Landsat 9

**LMOC-36990:** The LMOC shall perform constraint checking on a command load planning period which overlaps a previous plan.
**Rationale:** Whenever building a load, the mission planning system shall be capable of recalling the previous plan in order to ensure constraints are applied correctly across the entire load, including the transition from one load to the next. The OPSCON is that there will be at least 24 hours of overlap between successful load builds.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5141:** The LMOC shall restrict mission data flow to observatory mass storage as necessary to enforce the constraint against the maximum data volume.
**Rationale:** In order to avoid potential anomalies on-board the observatory, the LMOC must ensure that image acquisition schedules and housekeeping data transfers to observatory mass storage do not violate the constraint on observatory mass storage data volume.
**Mission:** Landsat 8 and Landsat 9

**LMOC-22659:** The LMOC shall apply individual WRS-2 scene priority values when it is necessary to reduce image collection volume due to constraints in accordance to LTAP-9 (TBR).
**Rationale:** The DAM provides a priority value for each individual WRS-2 scene requested for acquisition; the LMOC receives this information from DPAS and applies it as part of the schedule generation process when absolutely necessary; this capability is generally only used during non-nominal operations to avoid violating observatory mass storage constraints. Note that priority values are individual to each scene and exist within both LTAP Collection Request and collection request inputs.
**Mission:** Landsat 8

**LMOC-5194:** The LMOC shall apply individual WRS-2 scene priority values when it is necessary to reduce image collection volume due to constraints in accordance to LTAP-9.
**Rationale:** The DAM provides a priority value for each individual WRS-2 scene requested for acquisition; the LMOC receives this information from DPAS and applies it as part of the schedule generation process when absolutely necessary; this capability is

generally only used during non-nominal operations to avoid violating observatory mass storage constraints. Note that priority values are individual to each scene and exist within both LTAP Collection Request and collection request inputs.
**Mission:** Landsat 9

**LMOC-18841:** The LMOC shall validate all mission planning products against mission resource constraints per mission phase.
**Rationale:** Mission Resource constraints arise from many sources including the observatory CARD, instrument operations procedures, ground system constraints, and constraints that operations decide to impose for operability reasons. These constraints will change through each phase of the mission and impact the mission operations differently. As such the constraints need to be applied to all products throughout the missions phases to ensure the continuity of operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3391:** The LMOC shall provide users with the ability to configure mission resource constraints per mission phase.
**Rationale:** Users need the ability to define, manage and configure individual mission resource constraints and quantify their impact for mission planning functions. Users will also link specific events, activities, or related constraint values to individual constraints as part of mission planning configuration. These constraints arise from many sources including the observatory CARD, instrument operations procedures, ground system constraints, and constraints that operations decide to impose for operability reasons.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3392:** The LMOC shall deconflict observatory activities based on a numerical observatory activity prioritization scheme.
**Rationale:** Automated deconfliction of activity schedules is facilitated by a relative activity priority scheme. A scheduling activity priority scheme within the LMOC will be used to allow certain activities to replace others (orbit maneuvers, calibrations, collection requests, LTAP Collection Request imaging). All imaging and calibration requests coming from DPAS are already assigned a numerical priority.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5324:** The LMOC shall allow authorized users to modify the observatory activity prioritization scheme.
**Rationale:** The high-level observatory activity prioritization scheme determined in conjunction with the Government will require changes in priority from time to time, especially when transitioning mission phases and in anomaly situations.
**Mission:** Landsat 8 and Landsat 9

### 3.5.6 Mass Storage Modeling

**LMOC-3400:** The LMOC mission planning capability shall model current and predicted observatory on-board data storage utilization.

**Rationale:** The LMOC must predict and monitor observatory on-board data storage utilization in order to ensure stored command loads won't violate observatory data storage constraints.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3401:** The LMOC mission planning capability shall model the observatory on-board data storage using observatory housekeeping telemetry input.
**Rationale:** The model should be updated to reflect current usage of the on-board recorder and any other storage to ensure future imaging activities can be scheduled within constraints.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3402:** The LMOC mission planning capability shall notify users of observatory on-board data storage modeling results via a report.
**Rationale:** Provide report of expected onboard data storage and any other storage utilization for planning and operational awareness.
**Mission:** Landsat 8 and Landsat 9

### 3.5.7 Contact Scheduling

**LMOC-22674:** The LMOC shall schedule S-Band contacts with a maximum contact separation of 200 minutes.
**Rationale:** Periodic contacts are needed every 200 minutes for mission data management and monitoring of S/C health and safety. This interval is defined by the desired approach to FOT/LMOC staffing, warning time for critical S/C faults, and nominal usage of the SSR (T&C contacts are required periodically to change the state of stored files to allow reuse of space for files that have been successfully received by the GS).
**Mission:** Landsat 8 and Landsat 9

**LMOC-22675:** The LMOC shall accommodate at least two S-band contacts per day with ten minutes or greater duration.
**Rationale:** This is necessary specifically to allow for uploading of large stored command loads.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3425:** The LMOC shall define each ground station along with the resource capabilities for that site.
**Rationale:** Define each ground station's capabilities within the LMOC system for S Band Uplink, S Band Downlink, X Band Real Time Downlink, X Band Playback Downlink (e.g. IC stations only receive X-band Real Time downlink).
**Mission:** Landsat 8 and Landsat 9

**LMOC-3423:** The LMOC shall provide users the ability to manage ground station definitions.

**Rationale:** Capability to modify existing station definitions, incorporate new ground stations, or delete stations that are no longer being used.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3426:** The LMOC shall provide users the ability to utilize any combination of contact resources defined for a ground station.
**Rationale:** Certain GNE stations can also be IC stations. The system must support configuration for and execution of GNE passes and IC passes for those particular stations and each individually. Also the capability needs to exist to plan and schedule any or all of the resources for the ground station. For example, this capability allows scheduling of GNE stations for S Band only, or X Band RT & PB only, or X Band RT only, and other selected-resource passes as determined by operations staff.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3427:** The LMOC shall designate each ground station as "Active" or "Inactive" for contact scheduling.
**Rationale:** Provides the ability to filter which ground stations will get incorporated into the schedule
**Mission:** Landsat 8 and Landsat 9

**LMOC-3406:** The LMOC shall generate predicted ground station contact/view periods for the observatory.
**Rationale:** LMOC needs to determine the contact times (AOS and LOS) for S and X Band links for each station.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3424:** The LMOC shall calculate all view periods for up to 50 ground stations.
**Rationale:** Need to plan and schedule view periods for GNE, NEN, SN and International Cooperator stations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3407:** The LMOC shall incorporate ground station antenna masks in computing predicted station contact/view periods.
**Rationale:** LMOC needs to ensure data is not transmitted while the ground station is obscured. The LMOC receives this information from IC and LGN stations for both S-Band, X-Band views.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3408:** The LMOC shall generate predicted SN contact/view periods for the observatory in accordance with the Space Networks User Guide.
**Rationale:** LMOC to determine when the TDRSS satellites are in view on the Landsat 9 observatory and when they may be scheduled for the highest available data rate.
**Mission:** Landsat 9

**LMOC-5265:** The LMOC shall generate predicted SN contact/view periods for the observatory per the LMOC to SN GS ICD.

**Rationale:** LMOC to determine when the TDRSS satellites are in view on the Landsat 8 observatory and when they may be scheduled for the highest available data rate.
**Mission:** Landsat 8

**LMOC-3409:** The LMOC shall support ground station scheduling of GNE resources in accordance with the GNE-LMOC ICD.
**Rationale:** The LMOC is responsible for observatory operations, a key function of which is optimizing the plan for and scheduling ground station resources.
**Mission:** Landsat 8 and Landsat 9

**LMOC-12582:** The LMOC shall generate CFRs for GNE per the GNE-LMOC ICD.
**Rationale:** The contact forecast request (CFR) provides a long range prediction of ground station contacts for each specific station. The LMOC passes a contact forecast report to the GNE ground stations for planning purposes.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3410:** The LMOC shall provide CFRs to the GNE per the GNE-LMOC ICD.
**Rationale:** The contact forecast request (CFR) provides a long range prediction of ground station contacts for each specific station. The LMOC passes a contact forecast report to the GNE ground stations for planning purposes.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3411:** The LMOC shall receive CFCs from the GNE per the GNE-LMOC ICD.
**Rationale:** The LMOC receives contact forecast confirmations (CFC) from each of the GNE ground stations for use in contact planning.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3412:** The LMOC shall schedule GNE contacts based on the CFR if a CFC is not received.
**Rationale:** The CFC is an optional response from GNE. It will only be transmitted to the LMOC in the event a conflict exists with the CFR.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5125:** The LMOC shall update the scheduled GNE contacts based on the CFC, if a CFC is delivered.
**Rationale:** The LMOC needs to update the scheduled contacts by what was confirmed from GNE in the CFC.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3413:** The LMOC shall process contact forecast confirmation (CFC) schedules issued by the GNE.
**Rationale:** Provide an electronic interface for incorporation of scheduled GNE events in the master schedule.
**Mission:** Landsat 8 and Landsat 9

**LMOC-6231:** The LMOC shall receive IC notifications from DPAS per the DPAS-LMOC ICD.
**Rationale:** The DPAS receives information like stations description, conflict notices, administrative messages, and problem reports post contact from the ICs. These information are forwarded to LMOC in accordance to the DPAS-LMOC ICD (ie. email) such that the flight operations team make appropriate changes in the IC operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-12581:** The LMOC shall generate CCSs for GNE per the GNE-LMOC ICD.
**Rationale:** The LMOC passes the confirmed contact schedule (CCS) to the GNE ground stations with each scheduling load.
**Mission:** Landsat 8 and Landsat 9

**LMOC-12580:** The LMOC shall generate IC CCSs for DPAS per the DPAS-LMOC ICD.
**Rationale:** The LMOC provides the station-specific Confirmed Contact Schedules (CCS) to the DPAS for distribution to each IC.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3415:** The LMOC shall regenerate and resend the CCS to affected ground stations each time a changed command load results in a change to the CCS.
**Rationale:** Need to notify the GNE and IC ground stations of planned changes to resources or contacts to ensure any stations are not caught unaware.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3414:** The LMOC shall provide CCSs to the GNE per the GNE-LMOC ICD.
**Rationale:** The LMOC passes the confirmed contact schedule (CCS) to the GNE ground stations with each scheduling load.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3416:** The LMOC shall provide CCSs to DPAS per the DPAS-LMOC ICD.
**Rationale:** The LMOC provides the station specifies Confirmed Contact Schedules (CCS) to the DPAS for distribution to each IC.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5266:** The LMOC shall schedule NEN resources in accordance with the NENUG.
**Rationale:** Provide an interface to network scheduling offices for receipt of network schedule and incorporation of scheduled network events in the master schedule for Landsat 8.
**Mission:** Landsat 8

**LMOC-5268:** The LMOC shall provide scheduling request information to the NEN as defined in the NENUG.
**Rationale:** The LMOC performs contact scheduling for Landsat 8 with NEN in accordance with defined NASA processes.
**Mission:** Landsat 8

**LMOC-5269:** The LMOC shall receive contact confirmation schedules from the NEN as defined in the NENUG.
**Rationale:** The LMOC receives contact scheduling confirmations for Landsat 8 from the NEN for use in contact planning.
**Mission:** Landsat 8

**LMOC-3417:** The LMOC shall schedule NEN resources in accordance with the NENUG.
**Rationale:** Provide an interface to network scheduling offices for receipt of network schedule and incorporation of scheduled network events in the master schedule for Landsat 9.
**Mission:** Landsat 9

**LMOC-3418:** The LMOC shall provide scheduling request information to the NEN as defined in the NENUG.
**Rationale:** The LMOC performs contact scheduling for Landsat 9 with NEN in accordance with defined NASA processes.
**Mission:** Landsat 9

**LMOC-3419:** The LMOC shall receive contact confirmation schedules from the NEN as defined in the NENUG.
**Rationale:** The LMOC receives contact scheduling confirmations for Landsat 9 from the NEN for use in contact planning.
**Mission:** Landsat 9

**LMOC-3420:** The LMOC shall process contact confirmation schedules issued by the NEN.
**Rationale:** Provide an electronic interface for incorporation of scheduled NEN events in the master schedule.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5271:** The LMOC shall schedule SN services in accordance with the LMOC to SN GS ICD.
**Rationale:** The LMOC uses the SNAS Client to communicate with the central SN scheduling entity for use in contact planning.
**Mission:** Landsat 8

**LMOC-18831:** The LMOC shall provide scheduling request information to the SN per the LMOC to SN GS ICD.
**Rationale:** The LMOC performs contact scheduling for Landsat 8 with the SN in accordance with defined NASA processes.
**Mission:** Landsat 8

**LMOC-18832:** The LMOC shall receive contact confirmation schedules from the SN per the LMOC to SN GS ICD.

**Rationale:** The LMOC receives contact scheduling confirmations for Landsat 8 from the SN for use in contact planning.
**Mission:** Landsat 8

**LMOC-3421:** The LMOC shall schedule SN services in accordance with the LMOC to SN GS ICD.
**Rationale:** The LMOC uses the SNAS Client to communicate with the central SN scheduling entity for use in contact planning.
**Mission:** Landsat 9

**LMOC-18834:** The LMOC shall provide scheduling request information to the SN per the LMOC to SN GS ICD.
**Rationale:** The LMOC performs contact scheduling for Landsat 9 with the SN in accordance with defined NASA processes.
**Mission:** Landsat 9

**LMOC-18835:** The LMOC shall receive contact confirmation schedules from the SN per the LMOC to SN GS ICD.
**Rationale:** Th LMOC receives contact scheduling confirmations for Landsat 9 from the SN for use in contact planning.
**Mission:** Landsat 9

**LMOC-3422:** The LMOC shall process contact confirmation schedules issued by the SN.
**Rationale:** Provide an electronic interface for incorporation of scheduled SN events in the master schedule.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3405:** The LMOC shall allow users to schedule and execute an S-band ground contact within 15 minutes of an in-view observatory contact.
**Rationale:** Capability for LMOC to support scheduling and execution of an emergency contact. This does not apply to IC ground contacts.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3404:** The LMOC shall schedule all contact activities within observatory and ground station operational resource constraints.
**Rationale:** LMOC needs to schedule contacts within the defined resources, constraints and availability of the mission (i.e. view periods, station status, antenna masks, etc.). Scheduled contacts need to be validated against constraints to ensure the reliability of the contact.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3429:** The LMOC shall schedule contact execution for all "active" ground stations within the specified planning period.
**Rationale:** Only include the active stations and their associated resources within the schedule.

**Mission:** Landsat 8 and Landsat 9

**LMOC-3428:** The LMOC shall schedule all "active" ground station contacts based upon station resource definition and selected station capabilities.
**Rationale:** The LMOC, as the manager of observatory activities, schedules ground station resources according to defined capabilities (i.e. GNE stations provide full capabilities for both X- and S-band, ICs are scheduled for X-band RT downlink only, SN only uses S-band, etc.). For GNE contacts, the LMOC also selects station capabilities for use during each contact (i.e. any combination of S-band, X-band RT, X-band RT+PB). For scheduling purposes it is important to clearly define which station capabilities will be utilized during each contact in the scheduling products exchanged with ground stations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3430:** The LMOC shall schedule a real-time X-band downlink of ground image data when within view of an active IC station.
**Rationale:** Support CONOPS to always downlink the RT X-band when ground imaging (i.e. descending, day, or night) is being performed over an IC. Calibration and engineering collections are not transmitted to the ICs.
**Mission:** Landsat 8 and Landsat 9

### 3.5.8 Activity and Request Scheduling

**LMOC-2461:** The LMOC shall generate and maintain the schedule of ground and observatory activities.
**Rationale:** The activity schedule need to be generated to including both ground and observatory activities over the course of the life of the mission. This activity schedule will include but is not limited to Image Data Collections, Collection Requests, Contact Scheduling, Mission Data Management, etc.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3432:** The LMOC shall plan and schedule all observatory housekeeping and maintenance activities.
**Rationale:** The LMOC is responsible for observatory health and safety.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3433:** The LMOC shall interface to the government provided L&EO timeline management tool for command and activity sequences.
**Rationale:** The LMOC interfaces with the tool to coordinate objectives, procedures, scripts, and command sequences (real-time and stored) needed to execute the L&EO and commissioning activities.
**Mission:** Landsat 9

**LMOC-3434:** The LMOC shall process planning system inputs bounded by user-specified start and stop times.

**Rationale:** Allows the user to control which portions of input data to incorporate (i.e. filtering out past events, etc.).
**Mission:** Landsat 8 and Landsat 9

**LMOC-3435:** The LMOC shall provide users the ability to manually incorporate event and activity requests into a schedule.
**Rationale:** Provide capability to interrupt potential automated operations and allow manual scheduling of activities.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3436:** The LMOC shall provide users the ability to manually modify an activity within the schedule.
**Rationale:** Allows authorized personnel to add / delete / modify activities through manual edit of the schedule of activities.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3437:** The LMOC shall automatically process all activity requests that fall into the current scheduling window when generating a schedule.
**Rationale:** Requires that when a schedule is being generated, all activity requests for the current scheduling period are used as input. (i.e. for a 96-hour window, any activity that falls into that 96 hour period will be automatically read or for a 72-hour window, any activity that falls into that 72 hour period will be automatically read.)
**Mission:** Landsat 8 and Landsat 9

**LMOC-36988:** The LMOC shall adjust imaging sequences to avoid PIB-T lockout conditions.
**Rationale:** Observatory Flight Software (FSW) aboard Landsat 9 & Landsat 8 requires the LMOC to adjust the record stop position to avoid the EOF SDU from being split across two CFDP PDUs.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2464:** The LMOC shall provide a display of planned, currently active and past observatory and ground activities.
**Rationale:** Display information so it is readily viewable within the LMOC and to external users.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3441:** The LMOC shall provide users the ability to configure the default activity schedule duration.
**Rationale:** The FOT and FSM will establish the optimal operations concept for each observatory, which can evolve over time. Landsat 9 can support longer schedules than Landsat 8.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3438:** The LMOC shall automatically check for activity conflicts, resource constraints and rule violations during schedule generation.

**Rationale:** The candidate schedule is always checked against an operator-defined set of rules concerning command sequencing / constraints.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3439:** The LMOC shall provide users the ability to be notified via display and log messages of conflicts, activity and resource constraint violations, and activity rule violations during planning and scheduling.
**Rationale:** Displays and reports are used to support the resolution of conflicts and violations. Log messages alert support personnel to the problem.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3440:** The LMOC shall generate a deconflicted activity schedule.
**Rationale:** LMOC to ensure there are no conflicts between any observatory activities.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2462:** The LMOC shall maintain a copy of the latest schedule of ground and observatory activities.
**Rationale:** The users must have functional access to the most recent schedule in the mission planning software to allow for further editing and troubleshooting.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2463:** The LMOC shall generate an activity report which lists potential observatory and ground activities within the current planning period.
**Rationale:** Descriptive information for upcoming events and activities is important to operations, and will be commonly displayed on wall mounted screens for viewing by wider LMOC audiences.
**Mission:** Landsat 8 and Landsat 9

### 3.5.9 Mission Data Management

**LMOC-5375:** The LMOC shall allow the user to select between the modes of observatory mass storage management.
**Rationale:** The observatories support two modes of MDM which is configurable by the FOT: a mode where the LMOC explicitly commands deletion of files (i.e. LMOC Managed Mode); and a mode where the LMOC only unprotects files and the s/c overwrites unprotected data as necessary (i.e. Spacecraft Overwrite Mode).
**Mission:** Landsat 8 and Landsat 9

**LMOC-6228:** The LMOC shall ensure delivery of at least 99.6% of the mission data acquired by the observatory to the ground, measured on a monthly basis.
**Rationale:** The intent is to safely deliver all mission data to the archive, but human and system errors sometimes occur. The LMOC must maintain positive control of contact schedules and downlink execution, including mission data file retransmissions, to ensure successful mission data delivery. The allowance for loss occurs most in the area of accidental and erroneous file deletion on-board the observatory.
**Mission:** Landsat 8 and Landsat 9

**LMOC-20432:** The LMOC shall ensure imaging intervals are confirmed on the ground within 11.5 hours of observations for 85% of the data.
**Rationale:** To meet user demand, especially in short turnaround instances such as emergency response, acquired image data and corresponding routine calibration data must be successfully received on the ground in a timely manner.
**Mission:** Landsat 8 and Landsat 9

**LMOC-32077:** The LMOC shall be able to generate a SIFMT within 30 minutes upon receipt of the GPR.
**Rationale:** The LMOC needs to generate the SIFMT within a timely manner to support the confirmation of imaging intervals on the ground.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5374:** The LMOC shall command creation of housekeeping data files in observatory mass storage.
**Rationale:** The spacecraft provides functionality for storing telemetry in observatory mass storage as housekeeping data files upon command. The LMOC will utilize this capability to receive stored telemetry through the X-band downlink.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5373:** The LMOC shall assign a Root File ID to each commanded observatory mass storage housekeeping data file created.
**Rationale:** The LMOC is in charge of naming Landsat 8 housekeeping data files in mass storage so it can track them through the system; this requirement encompasses file name determination by the LMOC and commanding of the observatory to apply the names.
**Mission:** Landsat 8

**LMOC-5234:** The LMOC shall assign a Root File ID to each scheduled imaging interval.
**Rationale:** Root file names distinguish imaging intervals on-board the Landsat 8 observatory. This requirement encompasses the determination by the LMOC of the appropriate Root File ID and commanding of the observatory to apply the Root File IDs.
**Mission:** Landsat 8

**LMOC-5235:** The LMOC shall minimize the reuse of each Root File ID.
**Rationale:** On Landsat 8 to minimize the chance of error, any given possible Root File ID should be re-used as far apart as possible.
**Mission:** Landsat 8

**LMOC-5236:** The LMOC shall assign Root File IDs to imaging intervals sequentially.
**Rationale:** The imaging intervals on Landsat 8 need to clearly map to the assigned root file identifiers when further processed downstream.
**Mission:** Landsat 8

**LMOC-5237:** The LMOC shall manage Root File IDs assigned to imaging intervals.

**Rationale:** The LMOC, for Landsat 8, needs to track and maintain the list of possible Root File IDs against the list of currently assigned Root File IDs (i.e. files still exist in observatory mass storage or scheduled execution hasn't occurred yet) in order to avoid naming conflicts.
**Mission:** Landsat 8

**LMOC-5241:** The LMOC shall generate observatory mass storage commands using interval/Root File ID designations.
**Rationale:** LMOC needs to manage files on board the Landsat 8 observatory in a way the spacecraft is able to understand.
**Mission:** Landsat 8

**LMOC-5372:** The LMOC shall assign a Root File Name to each commanded observatory mass storage housekeeping data file created.
**Rationale:** The LMOC is in charge of naming Landsat 9 housekeeping data files in mass storage so it can track them through the system; this requirement encompasses file name determination by the LMOC and commanding of the observatory to apply the names.
**Mission:** Landsat 9

**LMOC-3450:** The LMOC shall assign a Root File Name to each scheduled imaging interval.
**Rationale:** Root file names distinguish imaging intervals on-board the Landsat 9 observatory. This requirement encompasses the determination by the LMOC of the appropriate Root File Name and commanding of the observatory to apply the Root File Names.
**Mission:** Landsat 9

**LMOC-5370:** The LMOC shall minimize the reuse of each Root File Name.
**Rationale:** To minimize the chance of error and/or data loss due to either software or human error, identical names should be used as far apart in time as the current pace of operations, and the total name space, allows.
**Mission:** Landsat 9

**LMOC-5131:** The LMOC shall assign Root File Names to imaging intervals sequentially.
**Rationale:** The imaging intervals on Landsat 9 need to clearly map to the assigned root file identifiers when further processed downstream.
**Mission:** Landsat 9

**LMOC-3451:** The LMOC shall manage Root File Names assigned to imaging intervals and all other files stored in observatory mass storage.
**Rationale:** The LMOC, for Landsat 9, needs to track and maintain the list of possible Root File Names against the list of currently assigned Root File Names (i.e. files still exist in observatory mass storage or scheduled execution hasn't occurred yet) in order to avoid naming conflicts.
**Mission:** Landsat 9

**LMOC-3452:** The LMOC shall generate observatory mass storage commands using interval/Root File Name designations.
**Rationale:** The LMOC needs to manage files on board Landsat 9 in a way the spacecraft is able to understand.
**Mission:** Landsat 9

**LMOC-3457:** The LMOC shall provide users the ability to select data in the observatory mass storage for downlink.
**Rationale:** This capability allows the FOT to determine which observatory mass storage files (Mission Data & Housekeeping Data) or locations to downlink and retransmit, and when. It includes LMOC system capabilities for control of spacecraft playback functionality, such as: enabling/disabling of autonomous file playback, file playback based on ground-uploaded schedule and playback of a contiguous range of observatory mass storage block addresses.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5379:** The LMOC shall be able to autonomously command all observatory mass storage file playback/transmission in real-time.
**Rationale:** In nominal operations the intent is to rely upon FSW automation of this capability to simplify command loads and command load generation. In certain mission phases such as commissioning, or during non-nominal operations, the LMOC may assume complete control of these functions.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5381:** The LMOC shall be able to autonomously command all observatory mass storage file playback/transmission through stored command loads.
**Rationale:** In nominal operations the intent is to rely upon FSW automation of this capability to simplify command loads and command load generation. In certain mission phases such as commissioning, or during non-nominal operations, the LMOC may need to specify which files are transmitted from the Observatory for a specific contact.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3447:** The LMOC shall command mission data files downlinked in real time over a GNE ground station to be marked as played back via stored command load.
**Rationale:** The L9 observatory has the capability to mark mission data files transmitted to the ground in real-time as played back, or not, based on stored command load. Mission data files are only marked as played back if they are completely transmitted to ground, and this functionality is desirable so that files which were already transmitted in real-time aren't also played back during the next ground station contact.
**Mission:** Landsat 9

**LMOC-3446:** The LMOC shall provide users the ability to designate imaging intervals as protected within the observatory mass storage.
**Rationale:** Ensure data is not overwritten on board before it has been downlinked.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5378:** The LMOC shall have the ability to modify the playback flag of observatory mass storage files.
**Rationale:** While this is not needed nominally, certain situations call for modification of the playback flag to prevent redundant playback of data (e.g. preemptive writing of the playback flag for mission data files successfully transmitted to a GNE station in real-time at the end of each contact to prevent redundant playback of data).
**Mission:** Landsat 8 and Landsat 9

**LMOC-3458:** The LMOC shall automatically command the retransmission of observatory mass storage files which have not been successfully received by the ground.
**Rationale:** Once the LMOC receives mission data file acknowledgements from GNE via the GNE Post-Pass Report (GPR), it can determine which mission data files must be retransmitted and execute the action. Housekeeping data file assessment prior to issuing any needed retransmit commands is ultimately the responsibility of the LMOC.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5371:** The LMOC shall determine the integrity of housekeeping data files received by the ground from observatory mass storage.
**Rationale:** The LMOC is in charge of confirming the data integrity of housekeeping data files received from observatory mass storage prior to executing appropriate management functions (e.g. delete, unprotect, retransmit). Checksums or acknowledgement accomplished through GNE processing can be used by the LMOC to confirm the data quality of the low-level packets.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3453:** The LMOC shall provide users the ability to command files in the observatory mass storage to be unprotected.
**Rationale:** This capability allows the FOT to manually command observatory management of observatory mass storage files (Mission Data & Housekeeping Data); it allows the observatory to overwrite files which are no longer protected. A manual FOT-managed approach may be used early in commissioning before activating automation.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3454:** The LMOC shall automatically command the unprotection of observatory mass storage files based on the status received by the GNE.
**Rationale:** Once the LMOC receives mission data file acknowledgements from GNE via the GNE Post-Pass Report (GPR), it can automatically allow acknowledged mission data files to be overwritten by the observatory. Housekeeping data file acknowledgement prior to issuing the unprotect command is ultimately the responsibility of the LMOC. This functionality (unprotect) supports the observatory managed mode of observatory mass storage management operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3449:** The LMOC shall autonomously remove protected designations from files on the observatory mass storage based on integrity checks within the MOC.
**Rationale:** This capability allows the LMOC to leverage observatory management of SSR files; it allows the observatory to overwrite files which are no longer protected.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3455:** The LMOC shall provide users the ability to command data in the observatory mass storage to be deleted.
**Rationale:** This capability allows the FOT to command observatory mass storage files (Mission Data & Housekeeping Data) or locations to be deleted if necessary.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3456:** The LMOC shall automatically command the deletion of observatory mass storage files which have been successfully received on the ground.
**Rationale:** Once the LMOC receives mission data file acknowledgements from GNE via the GNE Post-Pass Report (GPR), it can automatically overwrite acknowledged mission data files. Housekeeping data file acknowledgement prior to deletion is ultimately the responsibility of the LMOC. This functionality (LMOC-commanded deletion) supports the LMOC managed mode of observatory mass storage management operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3459:** The LMOC shall process GNE post-pass reports (GPR).
**Rationale:** The GPR is provided by GNE and is critical to observatory mass storage management; at a minimum, the GPR provides the LMOC with acknowledgements of CFDP checksums for each mission data file. The LMOC responds to the GPR by assessing the acknowledgements and executing appropriate observatory mass storage management actions.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3460:** The LMOC shall automatically maintain the current status of observatory mass storage files based on the schedule, observatory real-time telemetry, and ground-based file integrity checks.
**Rationale:** This capability is the core enabler of observatory mass storage management functions such as deleting, unprotecting, or retransmitting observatory mass storage files to create storage space for upcoming activities. The LMOC uses the active schedule, observatory mass storage file listing from real-time telemetry, the GPR assessment, and ground-based assessment of housekeeping data files in order to fully confirm and comprehend the receipt of the scheduled data collections. This is achieved through understanding which data files have been successfully received on the ground, which files have been lost forever (if any), and ultimately using the schedule to create mappings of mission data files to intervals.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3461:** The LMOC shall generate a Station Contacts Execution Summary (SCES) per the DPAS-LMOC ICD.

**Rationale:** The SCES will provide operations with details of all station contacts that include: Scheduled contact duration vs actual duration, GPR latency, SSR utilization, and X-band utilization.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5145:** The LMOC shall provide the Station Contacts Execution Summary (SCES) to DPAS per the DPAS-LMOC ICD.
**Rationale:** The SCES will provide operations with details of all station contacts that include: Scheduled contact duration vs actual duration, GPR latency, SSR utilization, and X-band utilization.
**Mission:** Landsat 8 and Landsat 9

## 3.6 Flight Dynamics

### 3.6.1 Orbit Determination and Products

**LMOC-3465:** The LMOC shall generate a definitive ephemeris for each operations day.
**Rationale:** Definitive ephemeris to be generated for each day of observatory operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3464:** The LMOC shall generate the definitive ephemeris of the observatory based on the GPS orbit data available in housekeeping telemetry at an accuracy of better than 24m in each axis, 3 sigma.
**Rationale:** Processing of Landsat 9 GPS orbit data in telemetry to ensure accuracy required to support image processing.
**Mission:** Landsat 9

**LMOC-5231:** The LMOC shall generate the definitive ephemeris of the observatory based on the GPS orbit data available in housekeeping telemetry at an accuracy of better than 30m in each axis, 3 sigma.
**Rationale:** Processing of Landsat 8 GPS orbit data in telemetry to ensure accuracy required to support image processing.
**Mission:** Landsat 8

**LMOC-12761:** The LMOC shall generate observatory orbit information for GNE per the GNE-LMOC ICD.
**Rationale:** The LMOC generates acquisition data (TLE/IIRV) for the GNE ground stations in support of planned contacts.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3468:** The LMOC shall provide observatory orbit information to GNE per the GNE-LMOC ICD.
**Rationale:** The LMOC passes acquisition data (TLE/IIRV) to the GNE ground stations in support of planned contacts.
**Mission:** Landsat 8 and Landsat 9

**LMOC-12762:** The LMOC shall generate IC observatory orbit information for DPAS per the DPAS-LMOC ICD.
**Rationale:** The LMOC generates acquisition data for DPAS to be distributed to the ICs.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3469:** The LMOC shall provide IC observatory orbit information to DPAS per the DPAS-LMOC ICD.
**Rationale:** The LMOC passes acquisition data to the DPAS for distribution to the ICs.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5273:** The LMOC shall provide Landsat 8 observatory orbit information to the NEN as defined in the NENUG.
**Rationale:** The LMOC passes acquisition data to the NEN in support of planned Landsat 8 contacts.
**Mission:** Landsat 8

**LMOC-12764:** The LMOC shall generate observatory orbit information for the NEN as defined in the NENUG.
**Rationale:** The LMOC generates acquisition data for the NEN in support of planned Landsat 9 contacts.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3470:** The LMOC shall provide Landsat 9 observatory orbit information to the NEN as defined in the NENUG.
**Rationale:** The LMOC passes acquisition data to the NEN in support of planned Landsat 9 contacts.
**Mission:** Landsat 9

**LMOC-5276:** The LMOC shall receive TDRSS satellite position updates from the FDF as defined per the LMOC to FDF MOA.
**Rationale:** It is required as part of SN services support during pre-launch, through LEO and Commissioning phase. In addition, during normal operations, to utilize SN support for non-routine operations like orbit and attitude maneuvers, routine proficiency exercises, and as needed contingency contacts, LMOC requires TDRS ephemeris from FDF to predict S-band visibility windows.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5275:** The LMOC shall provide Landsat 8 observatory orbit information to the SN via the SNAS per the LMOC to SN GS ICD.
**Rationale:** The LMOC passes acquisition data to the SN in support of planned Landsat 8 contacts.
**Mission:** Landsat 8

**LMOC-13121:** The LMOC shall generate observatory orbit information for the SN per the LMOC to SN GS ICD.

**Rationale:** The LMOC generates acquisition data for the SN in support of planned contacts.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3471:** The LMOC shall provide Landsat 9 observatory orbit information to the SN via the SNAS as defined in the LMOC to SN GS ICD.
**Rationale:** The LMOC passes acquisition data to the SN in support of planned Landsat 9 contacts.
**Mission:** Landsat 9

**LMOC-5298:** The LMOC shall receive orbit products from the FDF per the LMOC to FDF MOA.
**Rationale:** For Landsat 9, FDF provides orbit data during L&EO until the Observatory GPS data is available and the LMOC orbit determination system has been checked out. In addition, for both Landsat 8 and 9, FDF provides orbit data during nominal operations and contingency operations in case Observatory data is unavailable.
**Mission:** Landsat 8 and Landsat 9

**LMOC-13124:** The LMOC shall generate orbit products for the FDF per the LMOC to FDF MOA.
**Rationale:** The MOA defines the standard products needed when interacting with the FDF.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5297:** The LMOC shall provide orbit products to the FDF per the LMOC to FDF MOA.
**Rationale:** The MOA defines the standard products and delivery methods for interacting with the FDF.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3475:** The LMOC shall process FDF provided orbit products for flight dynamics computations during LEO, maneuver planning, and contingency operations.
**Rationale:** Use FDF provided orbit data during L&EO until the observatory GPS data is available and the LMOC orbit determination system has been checked out, for maneuver planning or during contingency operations when observatory data is unavailable.
**Mission:** Landsat 8 and Landsat 9

**LMOC-32118:** The LMOC shall provide ephemeris data products to the Constellation Coordination System (CCS) per the LMOC to ESMO ICD.
**Rationale:** The LMOC provides ephemeris data products to the Constellation Coordination System as a mean to track the operational health of the constellation and individual missions, as well as share mission products and perform orbital analyses.
**Mission:** Landsat 8 and Landsat 9

**LMOC-18838:** The LMOC shall support the Government-furnished Space Situational Awareness software to support Conjunction Assessment (CA).
**Rationale:** The LMOC will utilize the software provided by the Government, (ex. SpaceNav) to augment the Navigation and Mission Design Branch Conjunction Assessment ability and overall risk mitigation maneuver planning.
**Mission:** Landsat 8 and Landsat 9

**LMOC-18840:** The LMOC shall generate ephemeris data products for the Government-furnished Space Situational Awareness software.
**Rationale:** Orbit products need to be generated to support utilization of the Government-furnished Space Situational Awareness software.
**Mission:** Landsat 8 and Landsat 9

**LMOC-18839:** The LMOC shall provide ephemeris data products to the Government-furnished Space Situational Awareness software in accordance with the LMOC-SpaceNav ICD.
**Rationale:** Orbit products need to be delivered to support utilization of the Government-furnished Space Situational Awareness software (ex. SpaceNav).
**Mission:** Landsat 8 and Landsat 9

**LMOC-35069:** The LMOC shall provide orbit maneuver products to the Government-furnished Space Situational Awareness software in accordance with the LMOC-SpaceNav ICD.
**Rationale:** LMOC-SpaceNav ICD defines the product and delivery methods for interactions with the Government-furnished Space Situational Awareness software.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3467:** The LMOC shall generate 7-day Landsat 9 ephemeris data to support Conjunction Assessment (CA) and avoidance maneuver activities per the CARA-LMOC ICD.
**Rationale:** Deliver observatory orbit data for subsequent analysis by the Joint Space Operations Center (JSpOC) via Flight Dynamics Analysis Branch (FDAB) CA service according to the CARA-LMOC ICD.
**Mission:** Landsat 9

**LMOC-3474:** The LMOC shall provide ephemeris data products to CARA per the CARA-LMOC ICD.
**Rationale:** CARA-LMOC ICD defines products and delivery methods for interactions with CARA. Includes predictive ephemeris with orbit maneuver information.
**Mission:** Landsat 9

**LMOC-12578:** The LMOC shall generate 7-day Landsat 8 ephemeris data to support Conjunction Assessment (CA) and avoidance maneuver activities per the RSPP to L8 ICD.

**Rationale:** Deliver observatory orbit data for subsequent analysis by the Joint Space Operations Center (JSpOC) via Navigation and Mission Design Branch (NMDB) CA service according to the RSPP to L8 ICD.
**Mission:** Landsat 8

**LMOC-5272:** The LMOC shall provide ephemeris data products to SpaceNav per the LSDS-1986 LMOC to SpaceNav ICD.
**Rationale: LSDS-1986** LMOC to SpaceNav ICD defines products and delivery methods for interactions with SpaceNav. Includes predictive ephemeris with orbit maneuver information.
**Mission:** Landsat 8

### 3.6.2 Orbit Maneuver Planning

**LMOC-3478:** The LMOC shall perform observatory maneuver planning.
**Rationale:** Capability is required to support higher level requirements for Science Accommodation and Calibration Requirements. Maneuver planning will have to be performed throughout the life of the mission.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3483:** The LMOC shall generate maneuver plans to transfer the observatory from the injection orbit to the operational orbit.
**Rationale:** The LMOC needs to be able to perform maneuver planning from the injection orbit to nominal operational orbit. This includes the capability plan maneuvers in order to execute under-flys of an operational satellite.
**Mission:** Landsat 9

**LMOC-38228:** The LMOC shall provide user configurable generation of orbit maneuver plans in support of all open-loop observatory maneuver modes.
**Rationale:** The Landsat 9 and Landsat 8 observatories are capable of being enabled to execute an open-loop maneuver mode, in which each thruster is fired for a specified duration with attitude being controlled by the reaction wheels.
**Mission:** Landsat 8 and Landsat 9

**LMOC-38294:** The LMOC shall provide user configurable generation of orbit maneuver plans in support of all closed-loop observatory maneuver modes.
**Rationale:** The Landsat 9 and Landsat 8 observatories are capable of being enabled to execute multiple closed-loop maneuver modes. Closed-loop mode-timer fires each thruster in the set, while off-pulsing to maintain attitude, then ACS automatically transitions out of the Delta-V Burn mode. Closed-loop accumulated fires each thruster in the set, while off-pulsing to maintain attitude, then each thruster on time is added to the accumulated on time until the commanded on time duration is reached.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3485:** The LMOC shall maintain an observatory ground track to WRS-2 grid within +/- 5 km cross-track at the descending node.

**Rationale:** Maintain orbit with respect to the WRS-2 grid.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3489:** The LMOC shall support observatory decommissioning activities.
**Rationale:** LMOC needs to be able to perform planning and execution of observatory decommissioning activities.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3487:** The LMOC shall determine and maintain calibration coefficients for the observatory thrusters.
**Rationale:** Maintaining the characterization of the thrust scale factor per thruster bank is needed to perform maneuver planning over the course of the mission.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3488:** The LMOC shall monitor and predict observatory propellant usage throughout the life of the mission.
**Rationale:** Track usage of propellant during nominal operations to ensure enough is available for de-orbit activities.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3486:** The LMOC shall automatically identify and report constraint violations during maneuver planning.
**Rationale:** To prevent the execution of a maneuver from violating constraints on the observatory.
**Mission:** Landsat 8 and Landsat 9

**LMOC-13127:** The LMOC shall generate orbit maneuver plans for FDF per the LMOC to FDF MOA.
**Rationale:** The MOA defines the standard products needed for interacting with the FDF.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3481:** The LMOC shall provide orbit maneuver plans to FDF per the LMOC to FDF MOA.
**Rationale:** The MOA defines the standard products and delivery methods for interacting with the FDF.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3480:** The LMOC shall plan and execute collision avoidance maneuvers.
**Rationale:** Collision avoidance maneuvers mitigate risk of close approach(s) as indicated by conjunction analysis and recommendation and as dictated by project management direction. Collision avoidance maneuvers will have to be performed throughout the life of the mission.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5277:** The LMOC shall receive conjunction assessment data from SpaceNav per the LMOC to SpaceNav ICD.
**Rationale:** The SpaceNav data needs to be delivered to the LMOC in order to perform conjunction analysis.
**Mission:** Landsat 8

**LMOC-12584:** The LMOC shall generate orbit maneuver products for CARA per the CARA-LMOC ICD.
**Rationale:** CARA products need to be generated in the appropriate format per the CARA-LMOC ICD to support conjunction assessment.
**Mission:** Landsat 9

**LMOC-3482:** The LMOC shall provide orbit maneuver products to CARA per the CARA-LMOC ICD.
**Rationale:** CARA-LMOC ICD defines products and delivery methods for interactions with CARA.
**Mission:** Landsat 9

### 3.6.3 Attitude Maneuver Planning

**LMOC-3383:** The LMOC shall perform observatory attitude maneuver planning.
**Rationale:** Attitude maneuvers are required for off-nadir imaging, calibrations, and orbit maneuvers which will need to be performed over the course of the life of the mission.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3384:** The LMOC shall generate attitude maneuver plans in support of all observatory attitude maneuver activities.
**Rationale:** The LMOC needs to generate plans for the execution of non-nadir pointing, attitude maintenance, calibration and imaging maneuver activities.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3385:** The LMOC shall specify and generate attitude maneuver sequences.
**Rationale:** Translate the attitude plans into attitude commands or a series of attitude command sequences.
**Mission:** Landsat 8 and Landsat 9

### 3.6.4 Flight Dynamics Performance Monitoring

**LMOC-3495:** The LMOC shall provide users the ability to perform real-time evaluation of attitude data using predicted attitude and on-board attitude estimation parameters available in the observatory housekeeping telemetry.
**Rationale:** To alert operations and image processing of potentially poor attitude.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3491:** The LMOC shall provide users the ability to generate attitude estimates from raw sensor data contained in the observatory housekeeping telemetry.

**Rationale:** Off-line process utilizing Level-0 housekeeping data - general capability.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5307:** The LMOC shall validate that the on-board attitude estimates meet mission accuracy requirements.
**Rationale:** Off-line process to verify integrity of the on-board attitude estimation process.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3492:** The LMOC shall provide users the ability to validate the on-board attitude within 30 minutes of receipt of stored housekeeping data.
**Rationale:** To alert operations and image processing of potentially poor on-board attitude estimates.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3494:** The LMOC shall generate a definitive attitude history file covering a user specified time period.
**Rationale:** Ground processing is used to verify integrity of on-board attitude estimates and provide a corrected attitude for possible use by data processing, etc.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2460:** The LMOC shall receive earth orientation data from the USNO.
**Rationale:** The LMOC uses USNO earth orientation parameters (ie. UTC/UT1 corrections and pole wander) for orbit determination planning activities.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5323:** The LMOC shall incorporate earth orientation data received from the USNO for orbit products.
**Rationale:** The LMOC uses USNO earth orientation parameters (ie. UTC/UT1 corrections and pole wander) to accurately transform orbit states between inertial and terrestrial coordinate systems.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3493:** The LMOC shall notify users if the on-board ephemeris estimates vary from the ground generated predicted ephemeris by a user configurable threshold.
**Rationale:** The FOT need to be able to detect excessive drift between the on-board ephemeris and ground predict.
**Mission:** Landsat 8 and Landsat 9

**LMOC-18836:** The LMOC shall provide users the ability to generate ephemeris estimates from raw sensor data contained in the observatory housekeeping telemetry.
**Rationale:** Off-line processing utilizing Level-0 housekeeping data to generate orbital ephemeris estimates.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5196:** The LMOC shall calculate predicted orbit state vectors at 96 hours with errors no larger than:420 meters (3s) radial4200 meters (3s) along-track480 meters (3s) cross-track
**Rationale:** The FOT need to ensure that the accuracy of the predicted ephemeris is sufficient to use for activity and orbit determination planning. This accuracy assumes a solar flux (F10.7) value of less than 215 (10-22 W/m2/Hz) and no maneuver activity during the propagation interval.
**Mission:** Landsat 9

**LMOC-5242:** The LMOC shall calculate predicted orbit state vectors at 72 hours with errors no larger than:420 meters (3s) radial4200 meters (3s) along-track480 meters (3s) cross-track
**Rationale:** The FOT need to ensure that the accuracy of the predicted ephemeris is sufficient to use for activity and orbit determination planning. This accuracy assumes a solar flux (F10.7) value of less than 215 (10-22 W/m2/Hz) and no maneuver activity during the propagation interval.
**Mission:** Landsat 8

## 3.7 Telemetry and Command

### 3.7.1 Project Database

**LMOC-5301:** The LMOC shall receive a validated Landsat 8 Observatory Command and Telemetry Database (C&TDB) and updates from the Landsat 8 Spacecraft Vendor.
**Rationale:** The Landsat 8 Spacecraft Vendor, Orbital ATK, is the source for creating the Landsat 8 Observatory (S/C + instruments) T&C database.
**Mission:** Landsat 8

**LMOC-3498:** The LMOC shall receive a validated Landsat 9 Observatory Command and Telemetry Database(C&TDB) and updates from the Landsat 9 Spacecraft Vendor.
**Rationale:** The Landsat 9 Spacecraft Vendor is the source for creating the Landsat 9 Observatory (S/C + instruments) T&C database.
**Mission:** Landsat 9

**LMOC-6235:** The LMOC shall generate and maintain the Project Database (PDB) for the life of the mission.
**Rationale:** The PDB will be the sole database that contains all observatory and ground telemetry and commanding values required for operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3499:** The LMOC shall ingest the Observatory C&TDB and updates provided by the Spacecraft vendor into the PDB.
**Rationale:** The LMOC will use the delivered Observatory Command and Telemetry Database, including updates, as the baseline within the PDB.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3502:** The LMOC shall use the PDB as the T&C source database for all command and telemetry functions within LMOC subsystems.
**Rationale:** Utilize a common source database for all functions within the LMOC.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3500:** The LMOC shall provide configuration management of the PDB for the life of the mission.
**Rationale:** Access and changes need to be controlled to LMOC command and telemetry database as well as the implemented PDB configuration used on all LMOC systems.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3501:** The LMOC shall implement approved command and telemetry database modifications within the PDB.
**Rationale:** Provide capability to make PDB modifications without receiving a new delivery from the SC Vendor. Changes may be needed within the PDB to cover additional items not within the delivered observatory T&C database, such as Pseudo & Ground mnemonics, or errors contained within the delivered observatory T&C database.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5321:** The LMOC shall support PDB configuration changes without impacting critical telemetry and command functions.
**Rationale:** The LMOC PDB function for each observatory needs to support operations seamlessly in the midst of needed updates and related development activities
**Mission:** Landsat 8 and Landsat 9

**LMOC-3504:** The LMOC shall identify and designate hazardous commands in the command database.
**Rationale:** Flagging of commands that could endanger the safety of personnel working on the observatory during I&T, launch prep & launch or cause significant damage to the observatory or facilities is crucial.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3503:** The LMOC shall provide users the ability to search and sort observatory command and telemetry values defined in the PDB.
**Rationale:** Allows for ease in defining displays, finding commands.
**Mission:** Landsat 8 and Landsat 9

### 3.7.2 Command

**LMOC-3509:** The LMOC shall receive table load updates from the Landsat 9 spacecraft and instrument vendors.
**Rationale:** Flight Software updates need to be delivered to the LMOC in accordance to the Landsat 9 observatory vendors flight software management plan by an approved method (i.e.. Email, CD, Flash Drive, etc.).
**Mission:** Landsat 9

**LMOC-5309:** The LMOC shall receive table load updates from the Landsat 8 spacecraft and instrument vendors.
**Rationale:** Flight Software updates need to be delivered to the LMOC in accordance to the Landsat 8 observatory vendors flight software management plan by an approved method (i.e.. Email, CD, Flash Drive, etc.).
**Mission:** Landsat 8

**LMOC-6238:** The LMOC shall deliver table loads to the controlled repository.
**Rationale:** Vendor delivered table loads need to be archived in the controlled repository.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3510:** The LMOC shall receive ground reference image data from the Landsat 9 spacecraft and instrument vendors.
**Rationale:** Provide capability for the spacecraft and instrument vendors to deliver the Landsat 9 spacecraft's specific reference image.
**Mission:** Landsat 9

**LMOC-5310:** The LMOC shall receive ground reference image data from the Landsat 8 spacecraft and instrument vendors.
**Rationale:** Provide capability for the spacecraft and instrument vendors to deliver the Landsat 8 spacecraft's specific reference image.
**Mission:** Landsat 8

**LMOC-6239:** The LMOC shall deliver the ground reference image to the controlled repository.
**Rationale:** the exported Observatory GRI and Vendor delivered GRI need to be archived in the controlled repository.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3511:** The LMOC shall perform command encryption that is compatible with the Landsat 9 observatory.
**Rationale:** The LMOC needs to perform NIST FIPS PUB 197 compliant AES-256 bit CCM command encryption and authentication implemented using a NIST FIPS PUB 140-2 certified device to communicate with the Landsat 9 Observatory.
**Mission:** Landsat 9

**LMOC-5244:** The LMOC shall perform command encryption that is compatible with the Landsat 8 observatory.
**Rationale:** The LMOC needs to perform Caribou Command Encryption to support communication with the Landsat 8 Observatory.
**Mission:** Landsat 8

**LMOC-3512:** The LMOC shall employ command encryption and authentication bypass (clear mode) methods for commanding.

**Rationale:** Needed for commanding of simulators and also for S/C encryption implementation.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3519:** The LMOC shall apply the LEO-T header format on commands transmitted to the ground stations or space network.
**Rationale:** Maintain format compatibility with existing ground and space network interfaces.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3507:** The LMOC shall implement the CCSDS command format employing Space Packet Protocol - CCSDS 133.0-B-1 Correction-2 per the L9 Space to Ground ICD.
**Rationale:** The Landsat 9 Space to Ground ICD will define the specific command uplink format and field values for CCSDS commands.
**Mission:** Landsat 9

**LMOC-23338:** The LMOC shall implement the CCSDS command format employing Communications Operations Requirements Procedure - CCSDS 232.1-B-1 per the L9 Space to Ground ICD.
**Rationale:** The Landsat 9 Space to Ground ICD will define the specific command uplink format and field values for CCSDS commands.
**Mission:** Landsat 9

**LMOC-23339:** The LMOC shall implement the CCSDS command format employing TC Space Data Link Protocol - CCSDS 232.0-B-1 per the L9 Space to Ground ICD.
**Rationale:** The Landsat 9 Space to Ground ICD will define the specific command uplink format and field values for CCSDS commands.
**Mission:** Landsat 9

**LMOC-23340:** The LMOC shall implement the CCSDS command format employing TC Synchronization and Channel Coding - CCSDS 231.0-B-1 per the L9 Space to Ground ICD.
**Rationale:** The Landsat 9 Space to Ground ICD will define the specific command uplink format and field values for CCSDS commands.
**Mission:** Landsat 9

**LMOC-5251:** The LMOC shall implement the CCSDS command format employing Space Packet Protocol - CCSDS 133.0-B-1 per the LDCM Space to Ground ICD.
**Rationale:** The LDCM Space to Ground ICD will define the specific command uplink format and field values for CCSDS commands.
**Mission:** Landsat 8

**LMOC-23349:** The LMOC shall implement the CCSDS command format employing Communications Operations Requirements Procedure - CCSDS 232.1-B-1 per the LDCM Space to Ground ICD.

**Rationale:** The LDCM Space to Ground ICD will define the specific command uplink format and field values for CCSDS commands.
**Mission:** Landsat 8

**LMOC-23350:** The LMOC shall implement the CCSDS command format employing TC Space Data Link Protocol - CCSDS 232.0-B-1 per the LDCM Space to Ground ICD.
**Rationale:** The LDCM Space to Ground ICD will define the specific command uplink format and field values for CCSDS commands.
**Mission:** Landsat 8

**LMOC-23352:** The LMOC shall implement the CCSDS command format employing TC Synchronization and Channel Coding - CCSDS 231.0-B-1 per the LDCM Space to Ground ICD.
**Rationale:** The LDCM Space to Ground ICD will define the specific command uplink format and field values for CCSDS commands.
**Mission:** Landsat 8

**LMOC-3508:** The LMOC shall generate commands to upload all observatory table types and formats defined in the Landsat 9 Command and Telemetry Handbook.
**Rationale:** The LMOC needs to be able to command the observatory to perform any of its operational capabilities and generate the correct load formats for tables or a portion of a table (i.e. memory). The handbook is provided as a deliverable item by the spacecraft bus vendor.
**Mission:** Landsat 9

**LMOC-5243:** The LMOC shall generate commands to upload all observatory table types and formats defined in the Landsat 8 Command and Telemetry Handbook.
**Rationale:** The LMOC needs to be able to command the observatory to perform any of its operational capabilities and generate the correct load formats for tables or a portion of a table (i.e. memory). The handbook is provided as a deliverable item by the spacecraft bus vendor.
**Mission:** Landsat 8

**LMOC-3513:** The LMOC shall construct real-time commands for uplink to the observatory.
**Rationale:** To perform real-time operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3514:** The LMOC shall construct command loads for uplink to the observatory.
**Rationale:** To execute pre-conceived activities on the observatory.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3443:** The LMOC shall convert to and from UTC and the observatory reference time.
**Rationale:** Allows conversion from observatory reference time to ground reference time (UTC) for LMOC products.

**Mission:** Landsat 8 and Landsat 9

**LMOC-22658:** The LMOC shall generate and transport a command from the T&C to the LMOC network edge within .5 seconds.
**Rationale:** Real-time telemetry and command (T&C) traffic latency impacts operations such as the command and control loop from the ground station antenna to the LMOC front-end system.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3444:** The LMOC shall time tag all stored commands (absolute and relative) with a resolution of no more than one (1) millisecond.
**Rationale:** Allows the operator to sequence and correlate ground system events. This is a format requirement and not an accuracy requirement. Mission specific values include:

L8 RTS/ATS - No milliseconds. 8 commands per second (FSW will auto stagger them)
L8 IWT – 20 millisecond resolution. Relative waits must be 20ms minimum, and ms waits between commands must be divisible by 20ms
L8 ROS – 20 millisecond resolution. Relative waits must be 20ms minimum, and ms waits between commands must be divisible by 20ms
L9 ATS – 100 millisecond resolution (must be exact multiple of 100ms relative to epoch)
L9 RTS – 100 millisecond resolution (must be exact multiple of 100ms)
**Mission:** Landsat 8 and Landsat 9

**LMOC-5120:** The LMOC shall be able to translate spacecraft and instrument table loads to command loads.
**Rationale:** Needed for the ability to format the received software updates from each observatories vendors for transition to the specific spacecraft.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5211:** The LMOC shall be able to build a command load using the Ground Reference Image.
**Rationale:** Needed in case the on-board addressable memory, including FSW, is invalid and needs to be imaged.
**Mission:** Landsat 8 and Landsat 9

**LMOC-18837:** The LMOC shall provide a tool that audits the generated observatory command loads with user defined mission constraints.
**Rationale:** All command loads need to be independently verified by an FOT tool that can apply additional constants, used during specific mission phases, before uplink to the spacecraft or simulators to ensure mission safety.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5214:** The LMOC shall deliver all command loads to the controlled repository.
**Rationale:** The developed command Loads need to be archived in the controlled repository.

**Mission:** Landsat 8 and Landsat 9

**LMOC-5127:** The LMOC shall be able to build spacecraft configuration tables that modify real-time telemetry content and sampling rates.
**Rationale:** The Landsat 9 observatory configuration tables will could be different based on the phase of the mission or during contingency activities.
**Mission:** Landsat 9

**LMOC-5135:** The LMOC shall be able to modify existing spacecraft configuration tables that modify real-time telemetry content and sampling rates.
**Rationale:** The Landsat 9 real-time telemetry content and sampling rates need the ability to be updated by the FOT.
**Mission:** Landsat 9

**LMOC-5136:** The LMOC shall be able to select between the spacecraft configuration tables that modify real-time telemetry content and sampling rates.
**Rationale:** The Landsat 9 observatory capabilities need to be selected to optimize operations given observatory bandwidth and communications.
**Mission:** Landsat 9

**LMOC-3516:** The LMOC shall maintain dedicated, single-console control of real-time command transmissions for each observatory at all times.
**Rationale:** This provides for a designated single command controller or command gateway (i.e. can allow or block transmission of commands submitted by another operator) to protect the observatory.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5118:** The LMOC shall support a command line interface for execution of spacecraft or ground commands.
**Rationale:** Provides capability to execute scripts, procs or commands to perform LMOC and Spacecraft activities (e.g. STOL).
**Mission:** Landsat 8 and Landsat 9

**LMOC-5190:** The LMOC shall verify commands prior to uplink by spacecraft ID.
**Rationale:** All commands, both real-time and via load, should be verified that they are intended for the specific observatory that is being commanded.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3505:** The LMOC shall provide a notification to the user prior to the use or transmission of hazardous commands.
**Rationale:** The LMOC needs to provide notification to operators if hazardous commands are included within a load file or are being transmitted to the spacecraft in real-time.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3515:** The LMOC shall support commanding in an autonomous (unattended) state.
**Rationale:** Provide the capability to run autonomous operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3517:** The LMOC shall provide command controller workstation event message access to all T&C workstations.
**Rationale:** Allows users in other areas to see event messages and logs from the real-time contact.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5207:** The LMOC shall deliver all command operations and command history, in raw (hex) and interpreted formats, to the controlled repository.
**Rationale:** All command messages issued to the observatory and time-tagged history of all commands sent to observatory needs to be archived.
**Mission:** Landsat 8 and Landsat 9

### 3.7.3 Telemetry

**LMOC-3522:** The LMOC shall process all housekeeping telemetry received.
**Rationale:** The LMOC may need to process real-time, playback and re-transmitted housekeeping telemetry from various S-band and/or X-band downlink virtual channels. The same packet may be received multiple times.
**Mission:** Landsat 8 and Landsat 9

**LMOC-22657:** The LMOC shall receive and process housekeeping telemetry from the LMOC network edge to the T&C within .5 seconds.
**Rationale:** Real-time telemetry and command (T&C) traffic latency impacts operations such as the command and control loop from the ground station antenna to the LMOC front-end system.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3527:** The LMOC shall process telemetry received form the ground stations at a maximum rate of 1.024Mbps.
**Rationale:** Maximum downlink rate of real-time telemetry from the spacecraft.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3524:** The LMOC shall process all spacecraft telemetry data with a resolution of one (1) millisecond.
**Rationale:** Allows the operator to sequence and correlate spacecraft events with a time resolution consistent with spacecraft time tagging of telemetry.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3537:** The LMOC shall process telemetry encapsulated with the LEO-T header format received from the ground stations or space network.

**Rationale:** Maintain format compatibility with existing ground and space network interfaces
**Mission:** Landsat 8 and Landsat 9

**LMOC-3521:** The LMOC shall implement processing of the CCSDS telemetry format employing Space Packet Protocol - CCSDS 133.0-B-1 Correction-2 per the L9 Space to Ground ICD.
**Rationale:** The Landsat 9 Space to Ground ICD will define the specific telemetry format and field values for CCSDS telemetry.
**Mission:** Landsat 9

**LMOC-23343:** The LMOC shall implement processing of the CCSDS telemetry format employing AOS Space Data Link Protocol - CCSDS 732.0-B-3 per the L9 Space to Ground ICD.
**Rationale:** The Landsat 9 Space to Ground ICD will define the specific telemetry format and field values for CCSDS telemetry.
**Mission:** Landsat 9

**LMOC-5245:** The LMOC shall implement processing of the CCSDS telemetry format employing Space Packet Protocol - CCSDS 133.0-B-1 per the LDCM Space to Ground ICD.
**Rationale:** The LDCM Space to Ground ICD will define the specific telemetry format and field values for CCSDS telemetry.
**Mission:** Landsat 8

**LMOC-23346:** The LMOC shall implement processing of the CCSDS telemetry format employing AOS Space Data Link Protocol - CCSDS 732.0-B-2 per the LDCM Space to Ground ICD.
**Rationale:** The LDCM Space to Ground ICD will define the specific telemetry format and field values for CCSDS telemetry.
**Mission:** Landsat 8

**LMOC-3528:** The LMOC shall process telemetry at all rates the spacecraft can generate as defined within the Landsat 9 Space to Ground ICD.
**Rationale:** LMOC needs to process telemetry from the spacecraft at different rates for SN contacts, and potentially multiple rates for ground station contacts.
**Mission:** Landsat 9

**LMOC-5253:** The LMOC shall process telemetry at all rates the spacecraft can generate as defined within the LDCM Space to Ground ICD.
**Rationale:** LMOC needs to process telemetry from the spacecraft at different rates for SN contacts, and potentially multiple rates for ground station contacts.
**Mission:** Landsat 8

**LMOC-3538:** The LMOC shall define derived-telemetry via user-defined equations utilizing all available telemetry sources.

**Rationale:** The LMOC requires derived-telemetry to cross telemetry sources (ie. using ground and flight telemetry) or within the same telemetry source to create a derived (pseudo) mnemonic.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3525:** The LMOC shall assess data quality of S-band telemetry.
**Rationale:** Received telemetry values are used to assess the health and status of the observatory, so the validity of the received values needs to be ensured.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5128:** The LMOC shall validate that the received telemetry matches the spacecraft configurations tables.
**Rationale:** All changes made to the Landsat 9 Observatory spacecraft configurations tables need to be verified to ensure that the expected change performs as expected with no unforeseen consequences
**Mission:** Landsat 9

**LMOC-3526:** The LMOC shall notify the user via event message when invalid telemetry is detected.
**Rationale:** Provide notification when bad telemetry is detected.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3529:** The LMOC shall record and distribute housekeeping telemetry packets received from up to two ground station sources simultaneously.
**Rationale:** LMOC needs to support hot-handovers between stations as well as receiving real-time from one and playback telemetry from another. Real time data is distributed to operators as per workstation configuration, playback data from the stations is recorded and distributed for ingest.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3535:** The LMOC shall replay housekeeping telemetry based on a user specified start and stop time and replay rate.
**Rationale:** LMOC to have the capability to playback telemetry through the system.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3536:** The LMOC shall allow users to replay Observatory On-Board Data Storage files at rates up to at least 2.6Mbps.
**Rationale:** Telemetry needs to be replayed and processed in a timely manner between contacts, to allow for further analysis of back orbit mnemonics and limits, without interfering real-time telemetry downlink.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5209:** The LMOC shall deliver all received real-time telemetry to the controlled repository.
**Rationale:** Real-time telemetry will need to be archived in the case that stored housekeeping data is corrupt or missing.

**Mission:** Landsat 8 and Landsat 9

**LMOC-38220:** The LMOC shall generate back-orbit event messages in real-time utilizing VC3 dump packets.
**Rationale:** The spacecraft can downlink all back-orbit event messages during a real-time contact for immediate analysis by the operations team. These event messages provide immediate insight to back-orbit activities and provides information to the operations team half an hour before X-band housekeeping would produce the same results.
**Mission:** Landsat 9

**LMOC-3531:** The LMOC shall provide stored housekeeping data to the Landsat 9 Spacecraft Vendor facility.
**Rationale:** The spacecraft vendor needs access to all stored telemetry to support sustaining operations.
**Mission:** Landsat 9

**LMOC-5311:** The LMOC shall provide stored housekeeping data to the Landsat 8 Spacecraft Vendor facility.
**Rationale:** The spacecraft vendor needs access to all stored telemetry to support sustaining operations.
**Mission:** Landsat 8

**LMOC-3532:** The LMOC shall deliver at least 99.1% of all housekeeping data acquired by the Landsat 8 observatory to the controlled repository, measured on a weekly basis.
**Rationale:** Ensure capture of all RAM SSOH data on S-Band and Playback HKDB data on X-Band for on-going trending, analysis, and Observatory Health and safety, measured on a weekly basis.
**Mission:** Landsat 8

**LMOC-5293:** The LMOC shall deliver at least 99.1% of all housekeeping data acquired by the Landsat 9 observatory to the controlled repository, measured on a weekly basis.
**Rationale:** Ensure capture of all stored and realtime housekeeping data on S-Band or X-Band for on-going trending, analysis, and Observatory Health and safety, measured on a weekly basis.
**Mission:** Landsat 9

### 3.7.4 Displays

**LMOC-3541:** The LMOC shall provide users the ability to display all processed housekeeping telemetry.
**Rationale:** The LMOC needs to be able to display all data produced by the observatory (e.g. for comparison between past and current values).
**Mission:** Landsat 8 and Landsat 9

**LMOC-3545:** The LMOC shall convert telemetry counts into Engineering Units (EUs) using predefined database conversion.
**Rationale:** Raw telemetry values need to be converted to standard units as defined within the database.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3546:** The LMOC shall convert discrete telemetry counts into user-configurable text or numeric values using predefined database conversion operators.
**Rationale:** Discrete telemetry points need to be converted to values that represent the state of the mnemonic, not just the numeric value.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3547:** The LMOC shall output telemetry counts in raw and converted format.
**Rationale:** Provide capability to view raw counts or converted values.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3540:** The LMOC shall generate and store user configurable textual and graphical data displays.
**Rationale:** To allow operators the opportunity to define data displays.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3549:** The LMOC shall assess the time since last update of displayed data and mark all parameters exceeding defined thresholds as stale in data quality.
**Rationale:** The viewer of data should know if data displayed has not updated in the expected time interval.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3548:** The LMOC shall display and plot user-selected telemetry data in real time.
**Rationale:** Provide capability to display values of telemetry or produce plots of real-time data.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3542:** The LMOC shall display file directory listings, tables, and dump data in human readable formats.
**Rationale:** Operators need to view and interpret tables and dumps to confirm contents.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5143:** The LMOC shall allow all displays to be configurable via Spacecraft ID.
**Rationale:** Provide capability to display values of telemetry or produce plots of real-time data by Spacecraft ID.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5215:** The LMOC shall support the display of a subset of housekeeping telemetry from each observatory on a single display.
**Rationale:** The Operator needs the ability to not have to reconfigure each workstations displays between different observatory contacts.

**Mission:** Landsat 8 and Landsat 9

**LMOC-3543:** The LMOC shall provide users the ability to export observatory table data.
**Rationale:** Allow effective data management, comparisons and distribution of the on-orbit and vendor delivered tables.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3544:** The LMOC shall provide users the ability to integrity check and verify the observatory table data.
**Rationale:** To ensure FSW on the observatory has not been corrupted and any changes are as expected by comparing the table dumps with the Ground Reference Image.
**Mission:** Landsat 8 and Landsat 9

### 3.7.5 Limit Checking

**LMOC-3551:** The LMOC shall perform limit checking of observatory telemetry.
**Rationale:** Limit checking to be performed on all types of telemetry received at the LMOC (real-time, stored state of health, diagnostics)
**Mission:** Landsat 8 and Landsat 9

**LMOC-5129:** The LMOC shall adjust limits, pages and constraints based on the selected spacecraft configuration table.
**Rationale:** Updates to the spacecraft configuration tables need to be reflected properly within operations.
**Mission:** Landsat 9

**LMOC-3552:** The LMOC limit checking function shall support Red-High (RH), Yellow-High (YH), Green, Yellow-Low (YL), and Red-Low (RL) limit ranges.
**Rationale:** Categorization of limit states.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3561:** The LMOC shall provide a user-configurable color coding scheme to display telemetry relative to limit ranges.
**Rationale:** Without specifying a particular approach, this requires an easily interpretable limit display to determine spacecraft health and safety, e.g. Red-High (RH), Yellow-High (YH), Green, Yellow-Low (YL), and Red-Low (RL). This is configurable to accommodate various forms of color-blindness.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3562:** The LMOC shall provide a user-selectable non-color-coded scheme to display telemetry relative to limit ranges.
**Rationale:** Identification of limit display using fonts, symbols, flashing text etc.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3553:** The LMOC shall provide users the ability to display telemetry limit states for all mnemonics.
**Rationale:** Display the limit range states when violations occur and when values are in-range.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3559:** The LMOC shall provide users the ability to enable or disable all limit checks.
**Rationale:** Capability to control limit checks on all mnemonics.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3558:** The LMOC shall provide users the ability to enable or disable groups of limit checks.
**Rationale:** Capability to control limit checks on groups of mnemonics.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3557:** The LMOC shall provide users the ability to enable or disable individual limit checks.
**Rationale:** Capability to control limit checks on an individual mnemonic basis.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3560:** The LMOC shall provide a command line interface to override limit definitions within the database.
**Rationale:** Provide the system the capability to re-define limit definitions or values temporarily, via user selection or scripted, without having to modify the contents of the PDB.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3555:** The LMOC shall monitor the configuration of the observatory and detect deviations from expected states.
**Rationale:** LMOC needs to monitor observatory configuration for health and safety.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3554:** The LMOC shall generate event messages when limit state transitions occur.
**Rationale:** Notify personnel of any limit violation.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3556:** The LMOC shall notify personnel of detected observatory configuration deviations.
**Rationale:** Alert LMOC personnel to unexpected states of the observatory.
**Mission:** Landsat 8 and Landsat 9

### 3.7.6 T&C Operations

**LMOC-3565:** The LMOC shall be able to support mission operations simultaneously with simulations.
**Rationale:** The LMOC system needs to be able to simultaneously perform T&C during nominal operations (ie. real-time commanding, real-time housekeeping telemetry receipt and ingest, stored housekeeping telemetry receipt and ingest, playback of internally recorded housekeeping telemetry) along with simulator T&C activities.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5144:** The LMOC shall have the ability to hot-swap multiple instances of the PDB to support commanding and telemetry functions.
**Rationale:** The T&C system(s) needs the ability to host multiple PDB versions or instances seamlessly in order to support development, I&T and operations activities.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3564:** The LMOC shall generate, log and display time-tagged event messages indicating all command activity, telemetry processing status, limit violations, spacecraft ID, configuration changes, and all error and warning conditions.
**Rationale:** Provides an on-going log of activities occurring during real-time contacts.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3566:** The LMOC shall provide users the ability to autonomously establish a command and telemetry link for all scheduled contacts.
**Rationale:** Autonomy needed for unattended operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3567:** The LMOC shall conduct handoffs between GNE, Near Earth Network (NEN), and Space Network (SN) with overlapping observatory view periods with loss of less than 10 seconds of data.
**Rationale:** Allows handoff from one ground station to another without significant loss of real-time telemetry (S-band link).
**Mission:** Landsat 8 and Landsat 9

**LMOC-3568:** The LMOC shall establish, monitor, and manage connections with GNE in accordance to the GNE-LMOC ICD.
**Rationale:** LMOC to configure links with GNE Ground stations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-36914:** The LMOC shall support TCP/IP encrypted commanding communication to the GNE per the GNE to LMOC ICD.
**Rationale:** The LMOC needs to support an TCP/IP encrypted commanding path to secure the unecrypted COMSEC Leading & Trailing Information as well as any non-COMSEC encrypted commanding.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3569:** The LMOC shall provide observatory commands to the GNE for S-band uplink in accordance to the GNE-LMOC ICD.
**Rationale:** The LMOC generates command loads for the execution of schedules and passes them to the observatory for command execution via the GNE ground stations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3570:** The LMOC shall receive real-time S-band housekeeping telemetry from the GNE in accordance to the GNE-LMOC ICD.
**Rationale:** The GNE ground stations stream downlinked S-Band real-time and playback telemetry data to the LMOC.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3571:** The LMOC shall receive S-band stored housekeeping telemetry from the GNE in accordance to the GNE-LMOC ICD.
**Rationale:** The GNE ground stations stream downlinked S-Band real-time and playback telemetry data to the LMOC.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3572:** The LMOC shall receive X-band stored housekeeping telemetry from the GNE in accordance to the GNE-LMOC ICD.
**Rationale:** The GNE ground stations capture X-Band stored state of health files and send to the LMOC.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3573:** The LMOC shall receive S-band housekeeping telemetry recorded and played back by the GNE in accordance with the GNE-LMOC ICD.
**Rationale:** The GNE ground stations stream telemetry data recorded at the stations to the LMOC.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3574:** The LMOC shall receive GNE post-pass reports (GPR) from the GNE in accordance to the GNE-LMOC ICD.
**Rationale:** The GNE creates post-pass reports containing mission data downlink status information and provides to the LMOC. The LMOC uses this information for metrics and mission data management.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5279:** The LMOC shall establish, monitor, and manage connections with NEN in accordance with the NENUG.
**Rationale:** LMOC to configure links with NEN Ground Stations.
**Mission:** Landsat 8

**LMOC-5280:** The LMOC shall provide observatory commands to the NEN for S-band uplink in accordance with the NENUG.
**Rationale:** The LMOC generates command loads for the execution of schedules and passes them to the Landsat 8 observatory for command execution via the NEN.

**Mission:** Landsat 8

**LMOC-5281:** The LMOC shall receive real-time Landsat 8 S-band housekeeping telemetry from the NEN in accordance with the NENUG.
**Rationale:** The NEN streams downlinked S-Band real-time and playback telemetry data to the LMOC. The Landsat 9 and Landsat 8 mission interfaces that need to be developed should be documented in appropriate ICD.
**Mission:** Landsat 8

**LMOC-5282:** The LMOC shall receive stored Landsat 8 S-band housekeeping telemetry from the NEN in accordance with the NENUG.
**Rationale:** The NEN streams downlinked S-Band real-time and playback telemetry data to the LMOC.
**Mission:** Landsat 8

**LMOC-5283:** The LMOC shall receive Landsat 8 S-band housekeeping telemetry recorded and played back by the NEN in accordance with the NENUG.
**Rationale:** The NEN streams telemetry data recorded at the stations to the LMOC.
**Mission:** Landsat 8

**LMOC-3575:** The LMOC shall establish, monitor, and manage connections with NEN in accordance with the NENUG.
**Rationale:** LMOC to configure links with NEN Ground Stations.
**Mission:** Landsat 9

**LMOC-3576:** The LMOC shall provide observatory commands to the NEN for S-band uplink in accordance with the NENUG.
**Rationale:** The LMOC generates command loads for the execution of schedules and passes them to the Landsat 9 observatory for command execution via the NEN.
**Mission:** Landsat 9

**LMOC-3577:** The LMOC shall receive real-time Landsat 9 S-band housekeeping telemetry from the NEN in accordance with the NENUG.
**Rationale:** The NEN streams downlinked S-Band real-time and playback telemetry data to the LMOC.
**Mission:** Landsat 9

**LMOC-3578:** The LMOC shall receive stored Landsat 9 S-band housekeeping telemetry from the NEN in accordance with the NENUG.
**Rationale:** The NEN streams downlinked S-Band real-time and playback telemetry data to the LMOC.
**Mission:** Landsat 9

**LMOC-3579:** The LMOC shall receive Landsat 9 S-band housekeeping telemetry recorded and played back by the NEN in accordance with the NENUG.
**Rationale:** The NEN streams telemetry data recorded at the stations to the LMOC.

**Mission:** Landsat 9

**LMOC-5285:** The LMOC shall establish, monitor, and manage connections with SN in accordance with the LMOC to SN GS ICD.
**Rationale:** LMOC to configure links with the Space Network.
**Mission:** Landsat 8

**LMOC-5286:** The LMOC shall provide Landsat 8 observatory commands to the SN for S-band uplink in accordance with the LMOC to SN GS ICD.
**Rationale:** The LMOC generates command loads for the execution of schedules and passes them to the observatory for command execution via the SN.
**Mission:** Landsat 8

**LMOC-5287:** The LMOC shall receive real-time Landsat 8 S-band housekeeping telemetry from the SN in accordance with the LMOC to SN GS ICD.
**Rationale:** The SN streams downlinked S-Band real-time data to the LMOC.
**Mission:** Landsat 8

**LMOC-5288:** The LMOC shall receive Landsat 8 S-band housekeeping telemetry recorded and played back by the SN ground station per the LMOC to SN GS ICD.
**Rationale:** The SN streams real-time telemetry data recorded at the stations to the LMOC.
**Mission:** Landsat 8

**LMOC-3581:** The LMOC shall establish, monitor, and manage connections with SN in accordance with the LMOC to SN GS ICD.
**Rationale:** LMOC to configure links with the Space Network.
**Mission:** Landsat 9

**LMOC-3582:** The LMOC shall provide Landsat 9 observatory commands to the SN for S-band uplink in accordance with the LMOC to SN GS ICD.
**Rationale:** The LMOC generates command loads for the execution of schedules and passes them to the observatory for command execution via the SN.
**Mission:** Landsat 9

**LMOC-3583:** The LMOC shall receive real-time Landsat 9 S-band housekeeping telemetry from the SN in accordance with the LMOC to SN GS ICD.
**Rationale:** The SN streams downlinked S-Band real-time data to the LMOC.
**Mission:** Landsat 9

**LMOC-3584:** The LMOC shall receive Landsat 9 S-band housekeeping telemetry recorded and played back by the SN ground station per the LMOC to SN GS ICD.
**Rationale:** The SN streams real-time telemetry data recorded at the stations to the LMOC.
**Mission:** Landsat 9

**LMOC-5126:** The LMOC shall receive Landsat 9 S-band stored housekeeping telemetry from the SN per the LMOC to SN GS ICD.
**Rationale:** Landsat 9 has the capability to send S-band stored housekeeping telemetry via the SN.
**Mission:** Landsat 9

**LMOC-3585:** The LMOC shall provide commands to the LMOC-CTP at the Landsat 9 S/C vendor facility.
**Rationale:** The LMOC-CTP interfaces between the spacecraft I&T equipment, and the LMOC, for certain spacecraft and observatory I&T activities.
**Mission:** Landsat 9

**LMOC-3586:** The LMOC shall receive telemetry from the LMOC-CTP at the S/C vendor facility.
**Rationale:** The Spacecraft I&T Facility streams downlinked Landsat 9 S-Band real-time and playback telemetry data to the LMOC during pre-launch testing.
**Mission:** Landsat 9

### 3.7.7 T&C Simulator

**LMOC-3588:** The LMOC shall interface with the LSIMSS GNE simulator.
**Rationale:** The LSIMSS low-moderate fidelity spacecraft simulator which includes GNE station emulation is Government-furnished to support the early stages of LMOC I&T. LMOC-LSIMSS interactions will include command and telemetry, and may include post-pass reports. LSIMSS software will be compatible for both Landsat 8 and Landsat 9 Missions.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5255:** The LMOC shall interface with the Landsat 8 Spacecraft / Observatory Simulator (SOS).
**Rationale:** The Spacecraft / Observatory Simulator is a High fidelity simulator that is Government-furnished. The Spacecraft / Observatory Simulator needs connectivity to communicate with the LMOC for interactions, which includes but is not limited to, command and telemetry and post-pass reports.
**Mission:** Landsat 8

**LMOC-5256:** The LMOC shall generate commands for transmission to the Landsat 8 Spacecraft / Observatory Simulator (SOS).
**Rationale:** For I&T, command check-out; command structure is the same as that of the observatory, however, the simulator does not support encryption.
**Mission:** Landsat 8

**LMOC-5257:** The LMOC shall process telemetry received from the Landsat 8 Spacecraft / Observatory Simulator (SOS)
**Rationale:** Telemetry format output form the simulator may be different than telemetry received from ground stations.

**Mission:** Landsat 8

**LMOC-5258:** The LMOC shall receive housekeeping telemetry from the Landsat 8 SOS via the Landsat 8 SOS-CTP.
**Rationale:** The SOS is the high-fidelity observatory simulator used for the more mature stages of I&T. The LMOC will be testing critical functionality against it.
**Mission:** Landsat 8

**LMOC-5259:** The LMOC shall provide commands to the Landsat 8 SOS via the Landsat 8 SOS-CTP.
**Rationale:** The SOS is the high-fidelity observatory simulator used for the more mature stages of I&T. The LMOC will be testing critical functionality against it.
**Mission:** Landsat 8

**LMOC-5150:** The LMOC shall interface with the Landsat 9 Spacecraft / Observatory Simulator (SOS).
**Rationale:** The Spacecraft / Observatory Simulator is a High fidelity simulator that is Government-furnished. The Spacecraft / Observatory Simulator needs connectivity to communicate with the LMOC for interactions, which includes but is not limited to, command and telemetry and post-pass reports.
**Mission:** Landsat 9

**LMOC-3589:** The LMOC shall generate commands for transmission to the Landsat 9 Spacecraft / Observatory Simulator (SOS).
**Rationale:** For I&T, command check-out; command structure is the same as that of the observatory, however, the simulator may not support encryption.
**Mission:** Landsat 9

**LMOC-3590:** The LMOC shall process telemetry received from the Landsat 9 Spacecraft / Observatory Simulator (SOS).
**Rationale:** Telemetry format output form the simulator may be different than telemetry received from ground stations.
**Mission:** Landsat 9

**LMOC-3591:** The LMOC shall receive housekeeping telemetry from the Landsat 9 SOS via the Landsat 9 SOS-CTP.
**Rationale:** The SOS is the high-fidelity observatory simulator used for the more mature stages of I&T. The LMOC will be testing critical functionality against it.
**Mission:** Landsat 9

**LMOC-3592:** The LMOC shall provide commands to the Landsat 9 SOS via the Landsat 9 SOS-CTP.
**Rationale:** The SOS is the high-fidelity observatory simulator used for the more mature stages of I&T. The LMOC will be testing critical functionality against it.
**Mission:** Landsat 9

**LMOC-3593:** The LMOC shall receive post-pass reports from the Landsat 9 SOS-CTP.
**Rationale:** The SOS-CTP will support post-pass reports in its SOS-specific configuration for I&T purposes.
**Mission:** Landsat 9

**LMOC-5152:** The LMOC shall interface with the Softsim Simulator.
**Rationale:** The Softsim is a moderate fidelity Landsat 9 software-based spacecraft simulator which that is Government-furnished. The Softsim needs connectivity to communicate with the LMOC for interactions, which includes but is not limited to, command and telemetry.
**Mission:** Landsat 9

**LMOC-3594:** The LMOC shall provide commands to the Softsim simulator.
**Rationale:** The Softsim Landsat 9 software-based spacecraft simulator is available prior to SOS delivery to the LMOC; it's primary purpose is I&T and product development at earlier stages in the lifecycle.
**Mission:** Landsat 9

**LMOC-3595:** The LMOC shall receive housekeeping telemetry from the Softsim simulator.
**Rationale:** The Softsim Landsat 9 software-based spacecraft simulator is available prior to SOS delivery to the LMOC; it's primary purpose is I&T and product development at earlier stages in the lifecycle.
**Mission:** Landsat 9

**LMOC-5216:** The LMOC shall interface with the Softbench Simulator.
**Rationale:** The Softbench is a moderate fidelity software-based Landsat 8 spacecraft simulator which that is Government-furnished. The Softbench needs connectivity to communicate with the LMOC for interactions, which includes but is not limited to, command and telemetry.
**Mission:** Landsat 8

**LMOC-5217:** The LMOC shall provide commands to the Softbench simulator.
**Rationale:** The Softbench Landsat 8 software-based simulator is used to support ongoing product development.
**Mission:** Landsat 8

**LMOC-5218:** The LMOC shall receive housekeeping telemetry from the Softbench simulator.
**Rationale:** The Softbench Landsat 8 software-based simulator is used to support ongoing product development.
**Mission:** Landsat 8

## 3.8 Test and Launch Support

**LMOC-3604:** The LMOC shall receive real-time telemetry from the Payload Processing Facility.
**Rationale:** The PPF provides Landsat 9 telemetry to the LMOC from the LMOC-CTP for pre-launch assessment and monitoring.
**Mission:** Landsat 9

**LMOC-12579:** The LMOC shall receive stored telemetry from the Payload Processing Facility.
**Rationale:** The PPF will provide Landsat 9 stored telemetry to the LMOC for pre-launch assessment and monitoring, as needed, during observatory performance testing.
**Mission:** Landsat 9

**LMOC-3607:** The LMOC shall forward real-time telemetry to the Commissioning Support Area.
**Rationale:** Provide Landsat 9 real-time telemetry to CSA for use by spacecraft vendor provided workstations.
**Mission:** Landsat 9

**LMOC-5113:** The LMOC shall provide all stored telemetry to the Commissioning Support Area (CSA) within 30 minutes of receipt at the LMOC
**Rationale:** Provide Landsat 9 stored housekeeping telemetry for use by the CSA to support LEO Operations.
**Mission:** Landsat 9

**LMOC-3601:** The LSR shall contain no more than 42 workstations for real-time telemetry displays and access to the LMOC trending and analysis system.
**Rationale:** Provide engineers supporting L&EO operations with the capability to view real-time telemetry.
**Mission:** Landsat 9

**LMOC-33873:** The LMOC shall provide 12 authorized users with policy based remote access to the LSR systems and LMOC websites.
**Rationale:** Remote access for authorized NASA users (e.g. NASA MSE's) is needed at the Spacecraft Vendor Facility by utilizing GFP via the USGS Bureau VPN to access eavesdropping line real-time and stored observatory telemetry, trending and analysis, engineering data, operational products and the LMOC document repository and ticketing system.
**Mission:** Landsat 9

**LMOC-3602:** The LMOC shall forward real-time telemetry to the LSR.
**Rationale:** During LEO, real-time telemetry is forwarded to the LSR to enable observatory monitoring by the launch support team. Pre-launch simulator (e.g. SOS, LSIMSS, Softsim) telemetry is forwarded to the LSR to support operations readiness activities.
**Mission:** Landsat 9

**LMOC-5112:** The LMOC shall forward the queued commands and messages from the Command Controllers Position to all LSR positions.
**Rationale:** Needed when verifying Landsat 9 commands during the Testing, Launch and Early Orbit phase of the mission by the launch support team.
**Mission:** Landsat 9

**LMOC-3597:** The Launch Support Room (LSR) shall have UTC clock and countdown clock displays visible from all console positions.
**Rationale:** Clocks and countdown displays are pivotal in alerting operations personnel and staff of key upcoming activities (i.e. launch, contacts).
**Mission:** Landsat 9

**LMOC-3598:** The LSR shall provide four wall mounted displays viewable from any workstation in the LSR.
**Rationale:** Heads-up status such as upcoming passes, trending data, telemetry displays may be best displayed to the entire LMOC including the LSR.
**Mission:** Landsat 9

**LMOC-3599:** The LSR shall provide users the ability to configure the information routed to the wall mounted displays.
**Rationale:** Personnel within the LSR will need to be able to sit at a centralized location and control a computer attached to a wall-mounted display (e.g. through the use of a Multi-in/out Matrix switch, wireless keyboards & mice or thin clients).
**Mission:** Landsat 9

**LMOC-3600:** The LSR shall provide a black/white and color printing capability in the LSR.
**Rationale:** Provide printing capability to engineers within the LSR.
**Mission:** Landsat 9

**LMOC-3608:** The mini-LMOC servers and equipment shall reside within 1, 24-U half rack.
**Rationale:** Define constraints for mini-LMOC size the spacecraft vendor needs to support.
**Mission:** Landsat 9

**LMOC-3609:** The mini-LMOC shall contain the T&C functionality needed to generate and transmit commands.
**Rationale:** Mini-LMOC needs to have basic command generation capabilities needed during Landsat 9 I&T.
**Mission:** Landsat 9

**LMOC-3610:** The mini-LMOC shall contain the T&C functionality needed to receive and process telemetry.

**Rationale:** Mini-LMOC needs to have basic telemetry processing capabilities to support Landsat 9 I&T.
**Mission:** Landsat 9

## 3.9 bLMOC

**LMOC-3612:** The LMOC shall provide a single string backup Landsat Multi-satellite Operations Center (bLMOC), to support space asset protection.
**Rationale:** Additional redundancy is not necessary in the backup capability to support a multiple failure scenario.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5203:** The bLMOC shall be able to accommodate simultaneous activities for both Landsat 8 and Landsat 9 observatories.
**Rationale:** The bLMOC space needs to allow for multiple spacecraft operations to occur at he same time without negatively impacting either mission.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3613:** The bLMOC shall maintain observatory health and safety, continue LTAP Collection Request collection and provide maneuver support for 30 days.
**Rationale:** Contingency operations need to be conducted out of the bLMOC until the primary LMOC has been restored or alternative plans are implemented. The bLMOC needs the capability to develop a contact schedule, perform MDM and to upload an LTAP Collection Request.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5164:** The bLMOC shall support long term trending and analysis, providing access to all telemetry received by the primary LMOC in the past year.
**Rationale:** The bLMOC needs to have a trending and analysis capability that can be utilized to support ongoing operations same as the LMOC.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3614:** The bLMOC shall support Telemetry and Commanding capabilities within 24 hours of the loss of the primary LMOC
**Rationale:** Operational control needs to be transferred to ensure space asset protection and to maintain observatory nominal operations, state of health evaluation, and observatory safety.
**Mission:** Landsat 8 and Landsat 9

**LMOC-38293:** The bLMOC shall have a combined availability of 98.5% for all subsystems.
**Rationale:** To ensure the bLMOC subsystems are available without the stringent constraints levied upon the primary LMOC T&C system. Availability is to be calculated as total downtime per month on a per-subsystem basis.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5165:** The bLMOC shall perform command encryption that is compatible with the Landsat 9 observatory.
**Rationale:** The bLMOC needs to perform NIST FIPS PUB 197 compliant AES-256 bit CCM command encryption and authentication implemented using a NIST FIPS PUB 140-2 certified device to communicate with the Landsat 9 Observatory.
**Mission:** Landsat 9

**LMOC-5261:** The bLMOC shall perform command encryption that is compatible with the Landsat 8 observatory.
**Rationale:** The bLMOC needs to perform Caribou Command Encryption to support communication with the Landsat 8 Observatory.
**Mission:** Landsat 8

**LMOC-5167:** The bLMOC shall schedule and execute GNE contacts to support observatory operations per the GNE-LMOC ICD.
**Rationale:** The bLMOC will utilize the GNE interfaces the same as the LMOC. This includes full T&C operations and all facets of observatory mass storage management.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5168:** The bLMOC shall schedule and execute NEN contacts to support Landsat 9 observatory operations per the NENUG.
**Rationale:** The bLMOC will utilize the NEN interfaces the same as the LMOC. This includes receiving NEN Post Pass Reports, playback files and S-Band Communications.
**Mission:** Landsat 9

**LMOC-5289:** The bLMOC shall schedule and execute NEN contacts to support Landsat 8 observatory operations per the NENUG.
**Rationale:** The bLMOC will utilize the NEN interfaces the same as the LMOC. This includes receiving NEN Post Pass Reports, playback files and S-Band Communications.
**Mission:** Landsat 8

**LMOC-5169:** The bLMOC shall schedule and execute SN contacts to support Landsat 9 observatory operations per the LMOC to SN GS ICD.
**Rationale:** The bLMOC will utilize the SN interfaces the same as the nominal LMOC. This includes receiving playback files, S-Band Communications and utilization of the SNAS Client.
**Mission:** Landsat 9

**LMOC-5290:** The bLMOC shall schedule and execute SN contacts to support Landsat 8 observatory operations per the LMOC to SN GS ICD.
**Rationale:** The bLMOC will utilize the SN interfaces the same as the nominal LMOC. This includes receiving playback files, S-Band Communications and utilization of the SNAS Client.
**Mission:** Landsat 8

**LMOC-5163:** The bLMOC shall support MDM and Mission Planning capabilities within 24 hours of the loss of the primary LMOC.
**Rationale:** Basic LTAP Collection Request scheduling without collection requests, Maneuver Support and MDM will be needed within the bLMOC to support continuing operations and for validity of observatory mass storage data.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5177:** The bLMOC shall schedule all planned collection activities that do not violate constraints.
**Rationale:** The bLMOC will only eliminate scenes or intervals that violate constraints. Planned collection activities include LTAP Collection Request, image collection, collection requests, and calibration activities (i.e. both routine cals and cals instantiated through a collection request).
**Mission:** Landsat 8 and Landsat 9

**LMOC-5174:** The bLMOC shall be able to plan and execute maneuvers for each observatory.
**Rationale:** While utilizing the backup capability the observatory may need to perform nominal maintenance, attitude and maneuvers.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5264:** The bLMOC shall interface with FDF to perform mission planning activities per the LMOC to FDF MOA.
**Rationale:** All products and exchanges that occur between the LMOC and FDS will need to be included in the bLMOC, which include but are not limited to maneuver plans and orbit products.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5175:** The bLMOC shall interface with the Government-furnished Space Situational Awareness software to perform mission planning activities per the LMOC-SpaceNav ICD.
**Rationale:** All products and exchanges that occur between the LMOC and SpaceNav will need to be included in the bLMOC, which include but are not limited to maneuver plans, conjunction assessments and orbit products.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5193:** The bLMOC shall interface with DPAS to perform mission data accounting, image acquisition scheduling and mission planning activities per the DPAS-LMOC ICD.
**Rationale:** All products and exchanges that occur between the LMOC and DPAS will need to be included in the bLMOC, which include but are not limited to the IDSR, STS, LTAP Collection Request, SCES, WRS2TTT, collection requests, collection request status and Definitive Ephemeris.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5170:** The bLMOC shall schedule a real-time X-band downlink of ground image data when within view of an active IC station.
**Rationale:** Support CONOPS to always downlink the RT X-band when ground imaging is being performed over an IC. Calibration and engineering collections are not transmitted to the ICs.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5178:** The bLMOC shall be able to fail-back to the LMOC within 48 hours of the LMOC restoration.
**Rationale:** Relatively quick turnaround required to transition back to routine operations in the LMOC when it returns from an extended outage; this capability includes system and data replication from the backup capability to the LMOC (i.e. transfer of all data, documentation, system configurations and any other information necessary to support ongoing operations).
**Mission:** Landsat 8 and Landsat 9

**LMOC-3616:** The LMOC shall automatically replicate scripts, software, documentation, and databases to the bLMOC within 24 hours of an update to the LMOC.
**Rationale:** Ensure the bLMOC is up to date with the latest software and configurations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5366:** The bLMOC shall provide configuration management of bLMOC systems and software.
**Rationale:** Prevent unauthorized changes from being applied and affecting the bLMOC systems.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5166:** The bLMOC shall provide a web based centralized location for the management of replicated LMOC documentation.
**Rationale:** The bLMOC will have a website that can be utilized for the creation of LOP's, COP's and for storing additional LMOC & bLMOC specific documentation.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3617:** The bLMOC shall employ security controls of NIST PUB 800-53 Rev 4 moderate level system in accordance with corresponding USGS guidance within the Landsat 9 LMOC IT Security Control Matrix.
**Rationale:** The bLMOC IT security posture must meet NIST 800-53 Rev 4 for a moderate threat level system. The USGS also has corresponding guidance which must be adhered to as part of compliance with NIST 800-53.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5182:** The bLMOC shall restrict access to bLMOC systems to only authorized personnel.
**Rationale:** The bLMOC needs the ability to provide users with individual or group policy based access to bLMOC Systems.
**Mission:** Landsat 8 and Landsat 9

**LMOC-6233:** The bLMOC shall notify personnel of system faults and failures via a communications service.
**Rationale:** The intent is to notify operations personnel of faults or failures using a centralized and/or distributed system, as necessary via email, text, phone, or pager.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5262:** The bLMOC shall support testing with the Landsat 8 Spacecraft / Observatory Simulator (SOS).
**Rationale:** The SOS is the high-fidelity observatory simulator used for the more mature stages of I&T. The backup capability will be testing critical functionality against it before on-orbit operations.
**Mission:** Landsat 8

**LMOC-5171:** The bLMOC shall support testing with the Landsat 9 Spacecraft / Observatory Simulator (SOS).
**Rationale:** The SOS is the high-fidelity observatory simulator used for the more mature stages of I&T. The backup capability will be testing critical functionality against it before on-orbit operations.
**Mission:** Landsat 9

**LMOC-5173:** The bLMOC shall support testing with the Softsim simulator.
**Rationale:** The Softsim Landsat 9 software-based spacecraft simulator is available prior to SOS delivery to the LMOC; it's primary purpose is I&T and product development at earlier stages in the lifecycle.
**Mission:** Landsat 9

**LMOC-5226:** The bLMOC shall support testing with the Softbench simulator.
**Rationale:** The Softbench Landsat 8 software-based spacecraft simulator is used for product development.
**Mission:** Landsat 8

**LMOC-5179:** The bLMOC shall provide UTC clock and countdown clock displays with unobstructed visibility from the console positions.
**Rationale:** All spacecraft operations are referenced to UTC time.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5180:** The bLMOC shall provide a black/white and color printing capability within the bLMOC.
**Rationale:** Printers are required since hard copy of some items will be necessary.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5181:** The bLMOC shall utilize dual-pathed redundant uninterruptible power for all systems.
**Rationale:** Provide power redundancy to prevent localized power interruptions from affecting operations. This includes the use of Automatic Transfer Switch (ATS) for

supplying power to single Power Supply Units (PSU) devices like workstations and direct redundant Power Distribution Units (PDU) feeds to dual PSU devices like servers. Facility power outages and maintenance occur infrequently which can cause interruptions to bLMOC Operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3619:** The bLMOC shall support combined Landsat 8 and Landsat 9 operations within an area not to exceed 400 sq.ft.
**Rationale:** Define bLMOC operations room space allocations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3620:** The bLMOC servers and network equipment for both Landsat 8 and Landsat 9 shall reside within 4, 42-U racks.
**Rationale:** Constrain space needs within the bLMOC equipment room.
**Mission:** Landsat 8 and Landsat 9

**LMOC-33871:** The LMOC shall provide a contingency Landsat Multi-satellite Operations Center (cLMOC) at USGS EROS, to support space asset protection.
**Rationale:** Additional TIER 3/4 failure redundancy is necessary at the USGS Earth Resources Observation and Science Center to support backup operations capabilities should a regional or local impact to NASA Goddard Space Flight Center occur.
**Mission:** Landsat 8 and Landsat 9

**LMOC-33872:** The cLMOC shall support combined Landsat 8 and Landsat 9 operations within an area not to exceed 300 sq.ft.
**Rationale:** Define cLMOC operations room space allocations.
**Mission:** Landsat 8 and Landsat 9

## 3.10 Monitor and Trending

**LMOC-3623:** The LMOC shall provide system monitoring tools to track LMOC system performance.
**Rationale:** Supports health assessment of the LMOC and analysis in the event of failure.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3625:** The LMOC shall time tag all ground system event messages with the UTC time of event generation.
**Rationale:** Time-tag correlation is expected to be from spacecraft to ground (i.e. all times are converted to ground reference).
**Mission:** Landsat 8 and Landsat 9

**LMOC-22695:** The LMOC shall collect data quality, throughput, and volume metrics from LMOC subsystems and interfaces reflecting operations.

**Rationale:** Metrics describing data status at any given point in time are used to assess the state of operations, provide input to forward planning, and provide input to troubleshooting activities when necessary.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3624:** The LMOC shall log, track, and report system faults and failures.
**Rationale:** Applications report significant events to the centralized logging system and may maintain additional internal logging, etc. to support fault management
**Mission:** Landsat 8 and Landsat 9

**LMOC-3628:** The LMOC shall allow the user to enable, disable, filter, and display event messages.
**Rationale:** Provide capability to limit messages being displayed
**Mission:** Landsat 8 and Landsat 9

**LMOC-3626:** The LMOC shall notify personnel of system faults and failures via a communications service.
**Rationale:** The intent is to notify operations personnel of faults or failures using a centralized and/or distributed system, as necessary via email, text, phone, or pager.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3629:** The LMOC shall ingest and store all housekeeping telemetry for trending and analysis purposes.
**Rationale:** The LMOC may need to process real-time, playback and re-transmitted housekeeping telemetry from various sources. The same packet may be received multiple times.
**Mission:** Landsat 8 and Landsat 9

**LMOC-38423:** The LMOC shall be capable of re-ingesting archived housekeeping telemetry at nine times the realtime record rate.
**Rationale:** The LMOC may need to reprocess large quantities of archived housekeeping telemetry stored within the Controlled Repository to reconstitute the Trending and Analysis Database within a year.
**Mission:** Landsat 8

**LMOC-5346:** The LMOC shall provide a web-based interface to access the trending and analysis function.
**Rationale:** Local web-based access needs to exist to the trending and analysis system to support nominal operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5347:** The LMOC shall provide authorized users with policy based access to the trending and analysis web-based interface.
**Rationale:** Local web-based access needs to be specific per pre-defined policies (user or group) for on-demand trending access and any other functions.
**Mission:** Landsat 8 and Landsat 9

**LMOC-6108:** The LMOC shall quarantine any stored telemetry value marked as invalid.
**Rationale:** Data marked as invalid per analysis by the user shall not be used during telemetry analysis functions.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5197:** The LMOC shall be able to trend all stored telemetry data.
**Rationale:** All telemetry within the LMOC needs to be accessible by the system for trending and analysis purposes.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3643:** The LMOC shall produce a trending and analysis product containing up to 1,000,000 EU converted data points over a span of 24 hours within 5 minutes of request.
**Rationale:** Establish performance requirement for short term high resolution trending within the LMOC system using EU Converted data.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3634:** The LMOC shall trend up to ten different mnemonics per trend output.
**Rationale:** Display multiple parameters on one or more grids to support analysis of related parameters.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5320:** The LMOC shall display trend outputs from two observatories on a single display.
**Rationale:** Combine trending outputs for each observatory into a single display to support view and analysis of selected parameters.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3636:** The LMOC shall trend telemetry parameters in either raw or engineering unit (EU)-converted format.
**Rationale:** Provide capability to trend raw or converted values.
**Mission:** Landsat 8 and Landsat 9

**LMOC-32180:** The LMOC shall generate daily trending products within 1 hour of request meeting the following constraints: Minimum of 1400 mnemonics Minimum of 26 hour duration per mnemonic Minimum 6 minute statistical resolution per mnemonic
**Rationale:** Establish a performance requirement to support daily trending product generation using EU Converted data to support nominal operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3637:** The LMOC shall trend discrete telemetry parameters as numeric values.
**Rationale:** To be able to represent discrete states on a graph.
**Mission:** Landsat 8 and Landsat 9

**LMOC-33869:** The LMOC shall generate discrete conversion labels on trending plots as directed by the user.
**Rationale:** In conjunction with the capability to trend raw or converted values and being able to represent discrete states on a graph, labeling the axes of trend plots containing discrete values allows users to quickly and precisely interpret trending plots.
**Mission:** Landsat 8 and Landsat 9

**LMOC-33868:** The LMOC shall generate trend outputs which overlay consecutive user-defined time spans on a single plot.
**Rationale:** Allows multiple orbits of data to be plotted with the start of all orbits in the same location, enabling easy identification of telemetry behavior changes between orbits
**Mission:** Landsat 8 and Landsat 9

**LMOC-33870:** The LMOC shall generate statistical datasets consisting of a min, max, mean, and standard deviation value each orbit for the life of the mission.
**Rationale:** Allows rapid and precise analysis of multi-year trend sets for an observatory, particularly as related to subsystems which have consistent variations through each orbit.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3638:** The LMOC shall provide users the ability to set a sample filter rate for trended telemetry.
**Rationale:** Provide some operator control of output generation quantity.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3639:** The LMOC shall provide users the ability to trend basic statistical products including minimum, maximum, mean, and standard deviation values at a minimum.
**Rationale:** Allows feeding of standard statistical products through trending function; support long term trending.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3640:** The LMOC shall provide users the ability to both manually and automatically generate trending products and statistical products.
**Rationale:** Product generation can be under direct operator control, activity schedule or application triggers internally defined (i.e. Time of day or percent data collected, etc.).
**Mission:** Landsat 8 and Landsat 9

**LMOC-3630:** The LMOC shall generate trending products from telemetry that have been stored for trending and analysis purposes.
**Rationale:** Trend and analyze all recovered and/or stored engineering data.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3631:** The LMOC shall generate statistical products from telemetry that have been stored for trending and analysis purposes.
**Rationale:** Produce statistics on all recovered and/or stored engineering data.

**Mission:** Landsat 8 and Landsat 9

**LMOC-3632:** The LMOC shall generate trending and statistical products using user configurable product formats.
**Rationale:** Operators have the capability to specify formats for output (graphical, tabular, list, etc.).
**Mission:** Landsat 8 and Landsat 9

**LMOC-3633:** The LMOC shall generate trending and statistical products for any user-specified time interval.
**Rationale:** The function can be invoked over any time interval specified.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3635:** The LMOC shall generate trend products using user-selected time or parameter domain.
**Rationale:** Allows for time vs data and data vs data plots.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3641:** The LMOC shall be able to export any generated trending products or statistical products to a comma separated format.
**Rationale:** Personnel will need to import trending and statistical CSV products into other software or perform individual analysis.
**Mission:** Landsat 8 and Landsat 9

**LMOC-7583:** The LMOC shall generate trend and statistical products simultaneously for multiple users.
**Rationale:** Operators during pre-launch, commissioning and nominal operations will need to generate trending and statistical products at the same time.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3642:** The LMOC shall export ingested data in either raw or engineering unit (EU)-converted format.
**Rationale:** Export data in raw or converted values.
**Mission:** Landsat 8 and Landsat 9

**LMOC-33814:** The LMOC shall generate the KAL_ENCODER product per the DPAS-LMOC ICD.
**Rationale:** The KAL_ENCODER product is used by the cal/val community to understand the usage of the Landsat-8 TIRS instrument.
**Mission:** Landsat 8

**LMOC-33815:** The LMOC shall provide the KAL_ENCODER product per the DPAS-LMOC ICD.
**Rationale:** The KAL_ENCODER product is used by the cal/val community to understand the usage of the Landsat-8 TIRS instrument.
**Mission:** Landsat 8

## 3.11 Controlled Repository

**LMOC-3645:** The LMOC shall provide a centralized repository for storing and archiving all engineering data and operational products.
**Rationale:** There is value in management and consolidation of critical, commonly used data. All Engineering Data, Flight Operations Products and anything else needed for operations should be archived in a central location.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5160:** The LMOC shall provide a web-based interface to access the controlled repository.
**Rationale:** Local web-based access needs to exist to the controlled repository to support nominal operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5155:** The LMOC shall provide authorized users with policy based access to the controlled repository web-based interface.
**Rationale:** Local web-based access needs to be specific per pre-defined policies (e.g. user or group) for product access and any other functions.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5201:** The LMOC shall provide all engineering data and operational products to the controlled repository.
**Rationale:** Provide a controlled source where Engineering Data or Operational Products are available and may be retrieved or automatically delivered to the required subsystems.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5159:** The LMOC shall recognize when duplicate products are delivered to the system.
**Rationale:** During the life of the mission multiple files could be delivered and need to be flagged by the system as such.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5161:** The LMOC shall provide a user configurable option to handle any recognized duplicate products.
**Rationale:** FOT need to be able to re-deliver, accept or discard any files flagged as duplicate within the central repository.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5158:** The LMOC shall provide immediate access to all engineering data and operational products delivered within the last 7 days.
**Rationale:** LMOC subsystems and users need the ability to pull products directly from the controlled repository to support automated functionality and for general troubleshooting.

**Mission:** Landsat 8 and Landsat 9

**LMOC-7584:** The LMOC shall notify of engineering data and operational product availability upon delivery to the controlled repository.
**Rationale:** LMOC subsystems and users need to be notified of product deliveries to the controlled repository to support automated functionality and for support of nominal operations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3644:** The LMOC shall archive all engineering data and operational products for the life of the mission.
**Rationale:** Perform routine archival of engineering data products (ex. Trends, planning products, build information, reports, etc.) and operational products.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2411:** The LMOC shall receive and archive all housekeeping data for the life of the mission.
**Rationale:** Maintain archive of housekeeping data through all phases of the mission.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3518:** The LMOC shall archive all command operations and command history for the life of the mission.
**Rationale:** This archive includes all command messages issued to the observatory and provides a time-tagged history of all commands sent to observatory.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5315:** The LMOC shall provide access to any engineering data and operational products stored within the controlled repository.
**Rationale:** Personnel need to have the ability to search and retrieve any operational products throughout the life of the mission if it is stored in the repository.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2454:** The LMOC shall provide users the ability to generate human readable reports for products stored within the controlled repository.
**Rationale:** Ensure all historical products can be retrieved and transformed into a human readable product.
**Mission:** Landsat 8 and Landsat 9

**LMOC-2465:** The LMOC shall recover any engineering data and operational products stored within the controlled repository for the life of the mission.
**Rationale:** Maintain a record for all operational products with no loss or corruption of files throughout the life of the mission.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5205:** The LMOC shall provide configuration management of the engineering data and operational products within the controlled repository.

**Rationale:** Needed to restrict modifications to products and overall knowledge of the archives contents.
**Mission:** Landsat 8 and Landsat 9

## 3.12 Analytical Model

**LMOC-3649:** The LMOC shall provide an analytical modeling capability of the observatory and ground system interactions.
**Rationale:** Landsat program management will be making changes to GNE and other network configurations throughout the mission life. This capability will also be used to refine the operations concept and assess how to best deal with a degraded observatory in the event of failures.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3670:** The LMOC analytical modeling capability shall model in discrete time-steps.
**Rationale:** The modeling capability needs to address time and values with a relatively high amount of fidelity in order to achieve desired results.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3673:** The LMOC analytical model shall be capable of modeling one year of activities, including processing LTAP Collection Request and collection request inputs, within 90 minutes.
**Rationale:** The model needs to execute efficiently to allow for rapid evaluation of various scenarios and configurations. One year is a common modeling timeframe because of the significant seasonal changes driven by solar illumination, and campaign requests.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5119:** The LMOC shall provide a graphical interface to display the output of the analytical modeling.
**Rationale:** Needed by the user to visually represent the analytical modeling output after each run. The visual representation needs to convey the overall model for operations that was developed through its complete timeline.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3653:** The LMOC analytical model shall provide a log detailing the configuration settings used for each run.
**Rationale:** The model needs to be able to convey what the operations concept of the LMOC was during any given model run. For example, the command frequency and which observatory mass storage management process is employed or impacts from increasing or decreasing a particular ground station's utilization.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3669:** The LMOC analytical modeling capability shall provide users the ability to save input configurations for future use.

**Rationale:** Saving model input states is a useful function for re-running specific configurations on demand.
**Mission:** Landsat 8 and Landsat 9

**LMOC-5317:** The LMOC analytical modeling capability shall model mission planning and scheduling functions performed by the LMOC mission planning system algorithmically.
**Rationale:** In order to provide a strong foundation for high fidelity model outputs, actual LTAP Collection Requests and collection requests are input to the model, and corresponding TRUE-to-operations scheduling information must be integrated into the simulations.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3655:** The LMOC analytical modeling capability shall model and accept natively formatted LTAP Collection Request and collection request products as inputs.
**Rationale:** The model needs to evaluate collection requests, calibrations, and changes to the LTAP Collection Request within various operations concepts. As such, the model needs to ingest and process actual LTAP Collection Request products and collection requests.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3651:** The LMOC analytical modeling capability shall simulate interactions between the LMOC and observatory through the GNE and NEN ground stations.
**Rationale:** This allows LMOC operations concept changes to be modeled. The mechanics of how and when the LMOC interacts with the observatory, including command load frequency and station scheduling, can affect downlink and management of mission data and stored telemetry.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3650:** The LMOC analytical modeling capability shall simulate the interactions between GNE and NEN ground stations and the observatory.
**Rationale:** This allows network operations concepts and configurations to be traded. How the number/location of stations and the modes of stations (i.e. S-band and X-band, X-band only, S-band only) affect observatory mass storage management is a key focus for the modeling capability.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3667:** The LMOC analytical modeling capability shall model all combinations of ground station resources.
**Rationale:** The ground network operations concept is dynamic, and Landsat program management may change station operation mode. The potential also exists for station modes to be unavailable at times. Model modes need to include X-band RT only, X-Band PB only, X-Band RT+PB, S-band only, both X- and S-band.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3665:** The LMOC analytical modeling capability shall model the number of contacts per day at each ground station.
**Rationale:** The ground network operations concept is dynamic, and Landsat program management may add/remove stations and specify station usage on a per-pass basis.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3664:** The LMOC analytical modeling capability shall model ground station contact utilization.
**Rationale:** This includes contact time, and actual contact utilization (i.e. mission data volume or percentage of time spent transmitting mission data). These metrics are key to optimization of the overall ground station operations concept.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3654:** The LMOC analytical modeling capability shall model observatory command load frequency.
**Rationale:** How often the LMOC sends commands to the observatory is a key aspect of the LMOC operations concept.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3656:** The LMOC analytical modeling capability shall model observatory mass storage management.
**Rationale:** Observatory mass storage management activities are critical to LMOC operations. Activities include: reconciling observatory mass storage file directory listings against schedules and post-pass reports; commanding unprotection, deletion and/or retransmission of data within mass storage.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3657:** The LMOC analytical modeling capability shall model observatory mass storage utilization.
**Rationale:** Observatory mass storage is a key observatory resource which drives and/or constrains most of the station trades and operations concept changes.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3659:** The LMOC analytical modeling capability shall model stored state of health telemetry volumes on board the spacecraft and received by the LMOC.
**Rationale:** SOH management and latency is another key driver for GNE network configuration.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3660:** The LMOC analytical modeling capability shall model stored housekeeping telemetry latency from the observatory to the LMOC.
**Rationale:** Excessive latency on stored housekeeping telemetry puts the observatory at risk during contingencies.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3652:** The LMOC analytical modeling capability shall employ user specified configurations for ground station configuration and utilization, command load uplink frequency, and observatory mass storage management parameters.
**Rationale:** The model needs to be configurable to support trade studies, analyses, and various operations concepts.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3662:** The LMOC analytical modeling capability shall model mission data latency and volume received by the ground system.
**Rationale:** Some trades performed with a degraded observatory capability may involve optimization of science that can still be accomplished - for example if the observatory mass storage has a major loss of capability, there may be impacts to the overall ground network operations concept.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3674:** The LMOC analytical model shall model ground station contact utilization, observatory on-board data storage utilization and data latency with 90% accuracy when compared to actual on-orbit performance.
**Rationale:** The model needs to accurately reflect mission operations; on-board data storage and data latency accuracies include both mission data and telemetry.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3671:** The LMOC shall provide users the ability to export analytical modeling outputs in a human-readable format.
**Rationale:** The model outputs and results will be used by stakeholders.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3672:** The LMOC analytical modeling capability shall convey output values at each time-step for all outputs.
**Rationale:** Output information must be available at each point in time to achieve desired results.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3668:** The LMOC analytical modeling capability shall include ground station resources used in model outputs.
**Rationale:** Which modes ground stations are in can significantly impact the overall operations concept.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3666:** The LMOC analytical modeling capability shall include ground station utilization and contacts per day in model outputs.
**Rationale:** Understanding ground station utilization is a key objective of this modeling capability.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3658:** The LMOC analytical modeling capability shall include observatory mass storage utilization in model outputs.
**Rationale:** Understanding the observatory mass storage status at any point in time is a key objective of this modeling capability.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3661:** The LMOC analytical modeling capability shall include state of health volume and latency data in model outputs.
**Rationale:** Telemetry volumes and latencies provide input to overall ground station operations concepts.
**Mission:** Landsat 8 and Landsat 9

**LMOC-3663:** The LMOC analytical modeling capability shall include mission data compression ratio, latency, and volume received by the ground system in model outputs.
**Rationale:** Mission (science) data collection is the purpose of the mission. Understanding overall collection capability is paramount.
**Mission:** Landsat 8 and Landsat 9

# Appendix A Acronyms

| | |
|---|---|
| ACS | Attitude Control System |
| AM | Analytical Modeler |
| ANS | Alerts and Notifications |
| AOS | Acquisition of Signal |
| ATS | Automatic Transfer Switch |
| bLMOC | Backup Landsat Multi-Satellite Operations Center |
| C&DH | Command and Data Handling |
| C&TDB | Command and Telemetry Database |
| CA | Conjunction Assessment |
| CalCR | Calibration Collection Request |
| CARA | Conjunction Assessment Risk Analysis |
| CARD | Constraints and Restrictions Document |
| CCB | Configuration Control Board |
| CCS | Confirmed Contact Schedule or Constellation Coordination System |
| CCSDS | Consultative Committee for Space Data Systems |
| CFC | Contact Forecast Confirmation |
| CFDP | CCSDS File Delivery Protocol |
| CFR | Contact Forecast Request |
| CM | Configuration Management |
| CONOPS | Concept of Operations |
| COP | Contingency Operational Procedure |
| COTS | Commercial Off-the-Shelf |
| CPAW | Collection Planning and Analysis Workstation |
| CR | Change Request |
| CSA | Commissioning Support Area |
| CSV | Comma Separated Value |
| CTP | Command and Telemetry Processors |
| CVT | Cal/Val Team |
| DAM | Data Acquisition Manager |
| DHS | Department of Homeland Security |
| DM | Data Management |
| DNS | Domain Name System |
| DOI | Department of the Interior |
| DPAS | Data Processing and Archive System |
| DRC-16 | Design Reference Case (16 Days) |
| EOF | End of File |
| ESMO | Earth Science Mission Operations |
| EU | Engineering Units |
| FDAB | Flight Dynamics Analysis Branch |
| FDF | GSFC Flight Dynamics Facility |
| FDS | Flight Dynamics System |

| | |
|---|---|
| FIPS | Federal Information Processing Standards |
| FISMA | Federal Information Security Management Act |
| FOT | Flight Operations Team |
| FSM | Flight Systems Manager |
| FSW | Flight Software |
| GNE | Ground Network Element |
| GPR | GNE Post-Pass Report |
| GPS | Global Positioning System |
| GRI | Ground Reference Image |
| GS | Ground System |
| GSFC | Goddard Space Flight Center |
| GSIRD | Ground System Interface Requirements Document |
| GSVVP | Ground System Verification and Validation Plan |
| HVA | High Value Asset |
| I&T | Integration and Test |
| IC | International Cooperator |
| ICD | Interface Control Document |
| ID | Identifier |
| IDSR | Image Data Status Report |
| IIRV | Improved Inter-Range Vector |
| Inf | Infrastructure |
| IP | Internet Protocol |
| IRD | Interface Requirements Document |
| IT | Information Technology |
| JASON | Joint Activity Simulation for Observatories and Networks |
| JSpOC | Joint Space Operations Center |
| L8 | Landsat 8 |
| L9 | Landsat 9 |
| LDCM | Landsat Data Continuity Mission |
| LEO / L&EO | Launch and Early Orbit |
| LEO-T | Low Earth Orbiter-Terminal |
| LGN | Landsat Ground Network |
| LMO | Landsat Mission Operations |
| LMOC | Landsat Multi-Satellite Operations Center |
| LMOCRD | LMOC Requirements Document |
| LOP | Landsat Operational Procedure |
| LOS | Loss of Signal |
| LSIMSS | Landsat Scalable Integrated Multi-Mission Support System |
| LSR | Launch Support Room |
| LTAP | Long-Term Acquisition Plan |
| Mbps | Mega-bits per second |
| MDM | Mission Data Management |

| | |
|---|---|
| MOA | Memorandum of Agreement |
| MPS | Mission Planning System |
| MTASS | Multi-Mission Three-Axis Stabilized Spacecraft |
| NASA | National Aeronautics and Space Administration |
| NEN | Near Earth Network |
| NENUG | Near Earth Network Users' Guide |
| NIST | National Institute of Standards and Technology |
| NMDB | Navigation and Mission Design Branch |
| OBC | Onboard Computer |
| OLI | Operational Land Imager |
| OLI-2 | Operational Land Imager 2 |
| OPSCON | Operations Concept |
| OS | Operating System |
| PB | Playback |
| PDB | Project Database |
| PDU | Power Distribution Unit |
| PPF | Payload Processing Facility |
| PSU | Power Supply Units |
| PUB | Publication |
| RH | Red-High |
| RL | Red-Low |
| ROS | Relative Operational Sequence |
| RSPP | Robotic System Protection Plan |
| RT | Real Time |
| RTS | Relative Time Sequence |
| S/C or SC | Spacecraft |
| SAA | South Atlantic Anomaly |
| SCES | Station Contact Execution Summary |
| SCR | Special Collection Request |
| SDU | Service Data Unit |
| SEMP | Systems Engineering Management Plan |
| SIFMT | Scene-Interval-File Mapping Table |
| SN | Space Network |
| SNAS | Space Network Access System |
| SNUG | Space Network Users' Guide |
| SOH | State of Health |
| SOP | Standard Operating Procedure |
| SOS | Spacecraft / Observatory Simulator |
| SSOH | Stored State of Health |
| SSR | Solid State Recorder |
| STK | Systems Toolkit |
| STOL | System Test and Operations Language |
| STS | Scene Transmit Schedule |

| | |
|---|---|
| T&A | Trending & Analysis |
| T&C | Telemetry and Command |
| TBD | To Be Determined |
| TBR | To Be Resolved |
| TCP | Transmission Control Protocol |
| TDRS | Tracking and Data Relay Satellite |
| TDRSS | Tracking and Data Relay Satellite System |
| TIRS | Thermal Infrared Sensor |
| TIRS-2 | Thermal Infrared Sensor 2 |
| TLE | Two Line Element |
| UID | Unique Identifier |
| USGS | U.S. Geological Survey |
| USNO | U.S. Naval Observatory |
| UT1 | UTC Corrected |
| UTC | Coordinated Universal Time |
| WRS-2 | Worldwide Reference System-2 |
| WRS2TTT | Worldwide Reference System-2 to Time Translation Table |
| XML | Extensible Markup Language |
| YH | Yellow-High |
| YL | Yellow-Low |

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